Magnetic Monopoles, Duality, and Supersymmetry

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Abstract: These notes present a pedagogical introduction to magnetic monopoles and exact electromagnetic duality in supersymmetric gauge theories. They are based on lectures given at the 1995 Trieste Summer School in High Energy Physics and Cosmology and at the 1995 Busstepp Summer School at Cosener's House.

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0 Introduction

0.1 Introduction and Outline

The subject of magnetic monopoles has a remarkable vitality, resurfacing every few years with new focus. The current interest in magnetic monopoles centers around the idea of electromagnetic duality. Exact electromagnetic duality, first proposed in modern form by Montonen and Olive [1], has finally been put to non-trivial tests [2–7] in finite $N = 4$ Super Yang-Mills theory [8] and special finite $N = 2$ theories [9]. Although duality is far from being understood, the evidence is now sufficiently persuasive that the focus has turned from testing to duality to understanding its consequences and structure. Perhaps more significantly, it has also been understood that duality plays a central role in understanding strongly coupled gauge theories with non-trivial dynamics, particularly in their supersymmetric form [5, 10]. Here the duality is not exact but nonetheless the idea of a dual formulation of a strongly coupled theory in terms of weakly coupled magnetic monopoles is central and the dynamics of these theories is closely tied to the properties of magnetic monopoles, many of which can be studied semi-classically.

These lectures are intended to provide an introduction to the properties of magnetic monopoles which are most relevant to the study of duality. They have for the most part been kept at a level which should be appropriate for graduate students with a good grasp of quantum field theory and hopefully at least a passing acquaintance with supersymmetry and some of the tools of general relativity. There is also a recent review of exact electromagnetic duality by David Olive [11] which I highly recommend. There are many topics which are not covered, including, monopoles in gauge groups other than $SU(2)$, homotopy theory as applied to magnetic monopoles, the Callan-Rubakov effect, astrophysical implications of magnetic monopoles, experimental and theoretical bounds on the cosmic monopole abundance, and in general anything having to do with "real " magnetic monopoles as they might be found in nature. These omissions are more than made up for by the existence of excellent reviews which cover this material [12–15] and which the student should consult to complement the present lectures. I have also not covered most of the sophisticated mathematics related to the structure of the BPS monopole moduli space. A good reference for this material is [16].

What I have tried to do is to take a fairly direct route starting from the basics of magnetic monopoles and ending at the new evidence for S-duality found by A. Sen in February, 1994 [2]. These lectures are organized as follows. The first lecture begins with a brief discussion of some early examples of duality in physical systems. The basic ideas of electromagnetic duality and the Dirac monopole are then introduced followed by a

discussion of the 't Hooft Polyakov monopole. Lecture two begins a discussion of magnetic monopoles in the BPS limit of vanishing scalar potential. The Bogomol'nyi bound is derived and the BPS single charge monopole solution is presented. Spontaneous breaking of dilation symmetry and the Higgs field as "dilaton" are discussed. Collective coordinates are introduced through a concrete construction of the moduli space of the charge one BPS solution and then a more formal discussion of the moduli space of BPS monopoles is given. The third lecture discusses the dependence of monopole physics on the θ angle and introduces Montonen-Olive duality and its generalization to $SL(2,Z)$ known as S-duality. The coupling of fermions to magnetic monopoles is explored in the fourth lecture. Fermion zero modes are constructed and the effects of their quantization on the monopole spectrum is discussed. The fifth lecture explores the consequences of both $N = 2$ and $N = 4$ supersymmetry for monopole physics. The Bogomolnyi bound is revisited and related to a central extension of the supersymmetry algebra and the relation between BPS saturated states and short supermultiplets is briefly discussed. Finally, the basic features of supersymmetric quantum mechanics on the monopole moduli space are presented. The final lecture is devoted to the evidence for S-duality which comes from an analysis of the two-monopole moduli space following the work of Sen. I have included in this lecture a brief explanation of some elegant work of Manton's on the asymptotics of the two monopole moduli space. The final section contains some very brief remarks on open problems and recent developments. I have also taken the liberty of expanding some of the lectures beyond the material actually presented in order to provide what I hope will be a more useful review of duality.

There have of course been spectacular new developments in understanding duality in supersymmetric gauge theories with $N = 1, 2, 4$ supersymmetry and also in understanding duality in string theory [17, 18] which are not covered at all in these lectures. These developments show that electromagnetic duality is a profound new tool for probing the behavior of strong coupling dynamics. The material covered here is rather mundane in comparison but hopefully will provide students with some of the background necessary to appreciate and contribute to these new ideas.

0.2 Acknowledgements

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0.3 Conventions

We will use standard "field theory" relativity conventions with Minkowski space signature $(-+++)$ and $\epsilon^{0123} = +1$. Greek indices run over the range $0, 1, 2, 3$ while Roman indices run over the spatial indices $1, 2, 3$.

Generators T_a of a compact Lie algebra are taken to be anti-Hermitian. Gauge fields will be written as vector fields with an explicit gauge index A^a_μ, as Lie-algebra valued vector fields, $A_\mu = A^a_\mu T_a$, or as Lie-algebra valued one-forms $A = A_\mu dx^\mu$.

The speed of light c will always be set to 1. For the most part I will set $\hbar = 1$ and denote the gauge coupling by e. We will use Heaviside-Lorentz conventions for electromagnetism with factors of 4π appearing in Coulomb's law rather than in Maxwell's equations. The electric field of a point charge q is

$$\mathbf{E} = \frac{q\hat{\mathbf{r}}}{4\pi r^2}.$$

Similarly the magnetic field far outside a magnetic monopole of magnetic charge g is given by

$$\mathbf{B} = \frac{g\hat{\mathbf{r}}}{4\pi r^2}.$$

It is common in some monopole literature [14] to use definitions of the electric charge e and magnetic charge g which differ by a factor of 4π in order to preserve the quantization condition in the form originally given by Dirac, $eg = n/2$. Since the emphasis of these notes is on duality between electric and magnetic states, such a convention is inappropriate. Another common convention in the monopole literature [19] leaves $\hbar$ as an independent constant but sets the gauge coupling $e = 1$.

Other conventions involving supersymmetry and gamma matrices will be discussed in the text as they arise.

0.4 Exercises

Each lecture is followed by a set of exercises. Most of these are short and straightforward and are meant to reinforce the material covered rather than to seriously challenge the student. A few of the problems involve somewhat more advanced topics. As usual, serious students are strongly encouraged to work most of the problems.

1 Lecture 1

1.1 Duality

Saying that a physical system exhibits "duality" implies that there are two complementary perspectives, formulations, or constructions of the theory. To begin I will briefly describe duality in three systems which have had an impact on the search for duality in four-dimensional gauge theories.

In quantum mechanics we say that there is particle-wave duality meaning that quantum mechanically particles can exhibit wave like properties and waves (e.g. light) can exhibit particle like properties. We can think of this roughly speaking as the relation between the position space basis of states $\langle x|\psi\rangle$ and the momentum space basis $\langle p|\psi\rangle$ given by Fourier transform.

The harmonic oscillator provides a simple example of a system exhibiting "self-duality" in the sense that it looks the same in coordinate space and in momentum space. So consider

the harmonic oscillator Hamiltonian

$$H = \frac{p^2}{2m} + \tfrac{1}{2}m\omega^2 x^2 \tag{1.1}$$

with $[x, p] = i$. We can define a duality transformation which exchanges position and momenta by [1]

$$D : x \to p/m\omega, \qquad p \to -m\omega x \tag{1.2}$$

Note that this is a canonical transformation and thus preserves the commutation relations. Squaring D we find $D^2 = P$ with P the parity operator $P : x \to -x$. The fact that D is a symmetry of the harmonic oscillator is reflected in the fact that the ground state wave function and its Fourier transform are transformed into one another by the action of D, the Fourier transform of a Gaussian wave function is again Gaussian. Of course this is a rather trivial system, analogous to free field theory, and we will see in fact that the duality here is closely related to the electromagnetic duality of free Maxwell theory. We will later argue for an exact extension of electromagnetic duality in $N = 4$ super Yang-Mills theory which in some poetic sense should thus be regarded as the harmonic oscillator of four-dimensional gauge theory.

Another system, also exactly soluble, which exhibits a somewhat different kind of duality is the Ising model. This is defined by taking a set of spins σ_i taking the values ± 1 and living on a square two-dimensional lattice with nearest neighbor ferromagnetic interactions of strength J. The partition function at temperature T is

$$Z(K) = \sum_{\sigma} \exp(K \sum_{(ij)} \sigma_i \sigma_j) \tag{1.3}$$

where the sum on i, j runs over all nearest neighbors, the sum on σ over all spin configurations, and $K = J/k_B T$. This theory was solved explicitly by Onsager and exhibits a first-order phase transition to a ferromagnetic state at a critical temperature T_c. However even before Onsager's solution the critical temperature was computed by Kramers and Wannier using duality. They showed that the partition function (1.3) could be represented in two different ways as a sum over plaquettes of a lattice. In the first form the sum is over plaquettes of the original lattice with coupling K. In the second form one finds a sum over plaquettes of the dual lattice (the square lattice whose vertices are the centers of the faces of the original lattice) with coupling K^* where $\sinh 2K^* = 1/(\sinh 2K)$. Since the dual lattice is also a square lattice, the two formulations are equivalent, but with different values of K. Note also that high temperature ($K \ll 1$) or weak coupling is mapped to low temperature ($K^* \gg 1$) or strong coupling on the dual lattice. Now if the system is to have a single phase transition then it must occur at the self-dual point with $K = K^*$ or $\sinh(2J/k_B T_c) = 1$.

This model provides a more striking example of the use of duality. Duality provides non-trivial information about the critical behavior and relates a strongly coupled theory to a weakly coupled theory. Since many of the thorniest problems in theoretical physics involve

[1] This is clearly a discrete subgroup of a continuous symmetry which rotates p and x into each other.

strong coupling (e.g. quark confinement, high T_c superconductivity) it is very tempting to look for dualities which would allow us to use a dual weakly coupled formulation to do computations in such strongly coupled theories.

Another system which adds to this temptation occurs in two-dimensional relativistic field theory. The sine-Gordon model is defined by the action

$$S_{SG} = \int d^2x \left(-\tfrac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{\alpha}{\beta^2}\left(\cos\beta\phi - 1\right)\right). \tag{1.4}$$

This theory has meson excitations of mass $M_m = \sqrt{\alpha}$ and solitons which interpolate between different minima of the potential with mass $M_s = 8\sqrt{\alpha}/\beta^2$. By expanding the potential to quartic order we see that β^2 acts as the coupling constant for this theory. Thus the soliton mass is large (compared to the meson mass) at weak coupling.

Remarkably, this theory is known to be completely equivalent to an apparently quite different theory of interacting fermions knows as the Thirring model. The action of the Thirring model is

$$S_T = \int d^2x \left(-\bar\psi\gamma^\mu\partial_\mu\psi + m\bar\psi\psi - \frac{g}{2}\bar\psi\gamma^\mu\psi\,\bar\psi\gamma_\mu\psi\right) \tag{1.5}$$

At first sight these two theories appear completely different, but through the miracle of bosonization they are in fact completely equivalent [20, 21]. The map between the two theories relates the couplings through

$$\frac{\beta^2}{4\pi} = \frac{1}{1+g/\pi} \tag{1.6}$$

and maps the soliton of the SG theory to the fundamental fermion of the Thirring model and the meson states of the SG theory to fermion anti-fermion bound states. As in the Ising model, we see from (1.6) that strong coupling in one theory (i.e. large g) is mapped to weak coupling (small β) in the other theory. Thus duality provides a means of performing strong coupling calculations in one theory by mapping them to weak coupling calculations in a dual theory.

From these example we can extract certain general features of duality symmetries, although not all may be present in all examples. First, duality relates weak and strong coupling. Second, it interchanges fundamental quanta with solitons and thus exchanges Noether charges with topological charges. Finally it often involves a geometric duality, for example relating lattices to their duals. In four dimensional supersymmetric gauge theories we will find obvious generalizations of the first two features. The geometrical aspects of duality are also present, but only become clear when one considers general gauge groups.

The search for duality in four-dimensional gauge theories seems to have been motivated by the existence of dualities in these simpler systems, by electromagnetic duality and the results of Dirac, 't Hooft and Polyakov regarding the possible existence of magnetic monopoles, and by the work of Mandelstam, 't Hooft and others suggesting that confinement in QCD might arise as a dual form of superconductivity involving condensation of some sort of magnetically charged objects.

In spite of these hints, it has only been in the last two years that the idea of duality in non-trivial four-dimensional theories has been taken seriously by most particle physicists. These lectures will lead up to one non-trivial test of duality, but the skeptic could certainly remain unconvinced by the evidence discussed here. Although the conceptual underpinnings of duality remain quite mysterious, recent developments in gauge theory and string theory leave little room to doubt that duality exists and has significant applications.

1.2 Electromagnetic Duality

Maxwell's equations read

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \rho_e & \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} &= \mathbf{J}_e & \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0. \end{aligned} \tag{1.7}$$

When $\rho_e = \mathbf{J}_e = 0$ these equations are invariant under the duality transformation

$$D: \quad \mathbf{E} \to \mathbf{B}, \qquad \mathbf{B} \to -\mathbf{E}. \tag{1.8}$$

Note that D^2 takes $(\mathbf{E}, \mathbf{B}) \to (-\mathbf{E}, -\mathbf{B})$ which is a transformation by charge conjugation, C [2]. Thus the first thing we learn is that theories with exact duality must also be invariant under charge conjugation. The duality transformation (1.8) can be generalized to duality rotations parameterized by an arbitrary angle θ,

$$\begin{aligned} \mathbf{E} &\to \cos\theta \mathbf{E} + \sin\theta \mathbf{B}, \\ \mathbf{B} &\to -\sin\theta \mathbf{E} + \cos\theta \mathbf{B}. \end{aligned} \tag{1.9}$$

We will see later that this continuous duality transformation is broken to a discrete subgroup by instanton effects when duality is embedded in non-abelian gauge theories. If we write the Maxwell equations in covariant form in terms of the field strength $F^{\mu\nu}$ with $F^{0i} = E^i$ and $F^{ij} = \epsilon^{ijk} B_k$ then we have

$$\partial_\mu F^{\mu\nu} = -j_e^\nu, \qquad \partial_\mu * F^{\mu\nu} = 0 \tag{1.10}$$

where $*F^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\lambda\rho} F_{\lambda\rho}$ and the duality transformation (1.8) takes the form $F_{\mu\nu} \to *F_{\mu\nu}$. Note that in Minkowski space $** = -1$ in agreement with $D^2 = -1$.

The duality symmetry of the free Maxwell equations is broken by the presence of electric source terms. For this reason it is of no practical interest in everyday applications of electromagnetism. However the possibility that such a symmetry might nonetheless exist in some more subtle form has long intrigued physicists.

If we are to have such a symmetry it is clear that we will have to make the equations (1.7) symmetric by including magnetic source terms so that $\partial_\mu * F^{\mu\nu} = -k^\nu$ with k^ν the magnetic four-current. Of course in standard electromagnetism we actually take advantage of the lack of such source terms to introduce a vector potential. That is using $\partial_\mu * F^{\mu\nu} = 0$

[2] Note the analogy with the duality transformation (1.2) for the harmonic oscillator. This analogy can be made precise by decomposing the electromagnetic field in terms of normal modes.

we write $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ with $A^\mu = (\Phi, \mathbf{A})$ the vector potential. The association of a vector potential to a given field strength is not unique. The ambiguity is that of gauge transformations

$$A_\mu \to A_\mu - \partial_\mu \chi \tag{1.11}$$

which leave the field strength invariant.

Now recall that in coupling electromagnetism to quantum mechanics it is the vector potential A_μ and not just the field strength that plays a central role. Minimal coupling involves replacing the momentum operator by its covariant generalization

$$\mathbf{p} = -i\nabla \to -i(\nabla - ie\mathbf{A}) \tag{1.12}$$

with e the electric charge. The Schrödinger equation

$$i\frac{\partial \psi}{\partial t} = -\frac{1}{2m}(\nabla - ie\mathbf{A})^2\psi + V\psi \tag{1.13}$$

is then invariant under the combination of a gauge transformation on the vector potential and a phase transformation of the wave function:

$$\begin{aligned} &\psi \to e^{-ie\chi}\psi \\ &\mathbf{A} \to \mathbf{A} - \nabla\chi \equiv \mathbf{A} - \frac{i}{e}e^{ie\chi}\nabla e^{-ie\chi}. \end{aligned} \tag{1.14}$$

The latter form of the gauge transformation has been used to indicate that the fundamental quantity is the $U(1)$ group element $e^{ie\chi}$ and not χ itself.

Now returning to duality, we can ask, following Dirac, whether it is possible to add magnetic source terms to the Maxwell equations without disturbing the consistency of the coupling of electromagnetism to quantum mechanics. Dirac's argument [22], adapted to a modern perspective following the work of Wu and Yang [23] is given in the following section.

1.3 The Dirac Monopole à la Wu-Yang

Since we certainly do not believe that electromagnetism is correct down to arbitrarily small distance scales, let us first try to find a consistent description of a magnetic monopole excluding from consideration a region of radius r_0 around the center of the monopole. That is for $r > r_0$ we have a magnetic field

$$\mathbf{B} = \frac{g\hat{\mathbf{r}}}{4\pi r^2} \tag{1.15}$$

and we want to find a consistent description of quantum mechanics for $r > r_0$ in the presence of such a monopole magnetic field. Mathematically we are looking for a description in $R^3 - \{0\}$.

To couple a quantum mechanical charged particle to a background field we need the vector potential, but this seems inconsistent with having a magnetic monopole field. The solution involves making use of the ambiguity relating the vector potential to the field strength. To be specific, we can try to use different vector potentials in different regions as

long as the difference between them on overlap regions is that of a gauge transformation. Then the physically measurable field strength will be continuous and well defined. The simplest way to accomplish this is to divide a two-sphere S^2 of fixed radius $r > r_0$ into a Northern half N with $0 \leq \theta \leq \pi/2$, a Southern half S with $\pi/2 \leq \theta \leq \pi$ and the overlap region which is the equator E at $\theta = \pi/2$ (if desired the overlap region can be taken to be a band of finite width including the equator). The vector potential on the two halves is then taken to be [23]

$$\begin{aligned} \mathbf{A}_N &= \frac{g}{4\pi r}\frac{(1-\cos\theta)}{\sin\theta}\hat{\mathbf{e}}_\phi \\ \mathbf{A}_S &= -\frac{g}{4\pi r}\frac{(1+\cos\theta)}{\sin\theta}\hat{\mathbf{e}}_\phi. \end{aligned} \tag{1.16}$$

Note that on the two halves of the two-sphere the magnetic field as given by $\mathbf{B} = \nabla \times \mathbf{A}$ agrees with (1.15). Note also that $\mathbf{A}_{N,S}$ have singularities on (S, N) but are well defined in their respective patches.

Now to see if this construction makes sense we must check that the difference between $\mathbf{A}_N$ and $\mathbf{A}_S$ on the overlap region is indeed a gauge transformation. We have at $\theta = \pi/2$

$$\mathbf{A}_N - \mathbf{A}_S = -\nabla\chi, \qquad \chi = -\frac{g}{2\pi}\phi, \tag{1.17}$$

so that the difference is a gauge transformation. However the gauge function χ is not continuous. This was in fact inevitable as the following calculation of the enclosed magnetic charge demonstrates:

$$g = \int_N \mathbf{B}_N \cdot d\mathbf{S} + \int_S \mathbf{B}_S \cdot d\mathbf{S} = \int_E (\mathbf{A}_N - \mathbf{A}_S) \cdot d\mathbf{l} = \chi(0) - \chi(2\pi). \tag{1.18}$$

But physics does not require that χ be continuous. As is clear from (1.14), physical quantities will be continuous as long as $e^{-ie\chi}$ is continuous. This then gives us the condition $e^{-ieg} = 1$ or

$$eg = 2\pi n, \qquad n \in Z \tag{1.19}$$

which is the celebrated Dirac quantization condition [22].

Let me pause to make a few comments about what we have done so far.

1. As observed by Dirac, the presence of a single magnetic monopole anywhere in the universe is sufficient to guarantee that electric charge must be quantized. The quantization of electric charge is of course one of the fundamental experimental facts in particle physics and this provides an attractive explanation of why this should be the case.

2. The $U(1)$ group of gauge transformations has elements $e^{-ie\chi}$. If charge is quantized in units of some fundamental quanta e_1 then $\chi = 0$ and $\chi = 2\pi/e_1$ give the same gauge transformation. That is the range of the parameter χ is compact. It is useful to make the distinction between the compact one-parameter group which we will call $U(1)$ and the non-compact one-parameter group which we will call R which arises if charge is not quantized and thus the parameter range is the whole real line. Magnetic

monopoles require a compact $U(1)$ gauge group. Conversely, whenever a theory has a compact $U(1)$ gauge group it has magnetic monopoles. As we will see this includes grand unified theories where the $U(1)$ group is compact because it is embedded in a compact Lie Group such as $SU(2)$ but it also includes Kaluza-Klein theory where the $U(1)$ arises from symmetries of a compact space and string theory where the compactness seems necessary but is not yet completely understood in all cases.

3. Mathematically what we have done is to construct a non-trivial $U(1)$ principal fibre bundle. The base manifold is an S^2 at fixed radius which we cover with two coordinate patches. The fibers are elements of $U(1)$. The fibers are patched together with gauge transformations which are the transition functions. The magnetic charge is the first Chern class of this principal fibre bundle.

4. In the real world the low-energy gauge group includes more that just the $U(1)$ of electromagnetism, it includes $SU(3)$ of color and there are quarks which carry fractional electric charge. Must the Dirac condition be satisfied with respect to the electron or the quarks? The answer to this involves delving into monopoles in grand unified gauge theories. The brief answer is that, viewed from a distance, color is confined and monopoles must only satisfy the Dirac condition with respect to the electron. When viewed close-up such monopoles must also carry a color magnetic charge and the combination of the color magnetic charge and ordinary magnetic charge must satisfy a generalization of the Dirac quantization condition. For details see [24].

1.4 The 't Hooft-Polyakov Monopole

So far we have argued that there is a sensible quantum mechanics which includes magnetic monopoles as long as the charge is quantized and as long as we do not ask what happens inside the monopole. However there is in this framework no way to determine most of the properties of these monopoles including their mass, spin, and other quantum numbers. I now want to discuss a beautiful result of 't Hooft and Polyakov [25, 26] which allows us to probe inside the monopole and study its properties in detail.

Given that monopoles make sense if and only if the $U(1)$ gauge group is compact, it makes sense to look for them in theories where $U(1)$ is compact because it is embedded inside a larger compact gauge group [3]. The simplest possibility is the embedding $U(1) \subset SU(2)$ and it is this possibility which will occupy our attention for most of these lectures. We will take as a starting point the Yang-Mills-Higgs Lagrangian

$$\mathcal{L} = -\frac{1}{4}F^a_{\mu\nu}F^{\mu\nu}_a - \frac{1}{2}D^\mu\Phi^a D_\mu\Phi_a - V(\Phi) \tag{1.20}$$

where

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu - e\epsilon^a{}_{bc}A^b_\mu A^c_\nu \tag{1.21}$$

and the covariant derivative of Φ is

$$D_\mu\Phi^a = \partial_\mu\Phi^a - e\epsilon^a{}_{bc}A^b_\mu\Phi^c \tag{1.22}$$

[3] Compact $U(1)$ groups can also arise in Kaluza-Klein theory and in string theory and there one also finds magnetic monopole solutions [27–29].

with $a, b, c = 1, 2, 3$ labeling the adjoint representation of $SU(2)$. The potential $V(\Phi)$ is chosen so that the vacuum expectation value of Φ is non-zero. To be concrete we take $V(\Phi) = \lambda(\Phi^a\Phi_a - v^2)^2/4$.

By varying (1.20) with respect to A^a_μ and Φ^a we obtain the equations of motion

$$\begin{aligned} D_\mu F^{a\mu\nu} &= -e\epsilon^a{}_{bc}\Phi^b D^\nu \Phi^c \\ (D^\mu D_\mu \Phi)^a &= \lambda\Phi^a(\Phi^b\Phi_b - v^2). \end{aligned} \tag{1.23}$$

The Bianchi identity,

$$D_\mu * F^{a\mu\nu} = 0, \tag{1.24}$$

follows from the definition of $F^{a\mu\nu}$.

We will for the most part be interested in static solutions to the equations of motion (1.23). We will later include quantum effects by quantizing small fluctuations about such classical solutions.

It will also be useful in what follows to have an expression for the energy-momentum tensor for this theory. Straightforward computation gives [4]

$$\Theta^{\mu\nu} = F^{a\mu\rho}F_a{}^\nu{}_\rho + D^\mu\Phi^a D^\nu\Phi_a + \eta^{\mu\nu}\mathcal{L} \tag{1.25}$$

For $v = 0$, or for vanishing potential ($\lambda = 0$) the theory defined by (1.20) has a classical scale symmetry[5]. The conserved current is the dilation current $D_\mu = x^\nu\Theta_{\mu\nu}$ with

$$\partial_\mu D^\mu = \Theta^\mu{}_\mu = 0. \tag{1.26}$$

The case $V(\Phi) \equiv 0$ will occupy us later. We will argue that it still makes sense in this case to choose Φ to have an arbitrary but non-zero expectation value. This choice spontaneously breaks scale invariance. The resulting Nambu-Goldstone boson is traditionally called the dilaton, and is not to be confused with the "dilaton" field in string theory. Quantum mechanically scale invariance is broken by renormalization and the trace of the energy momentum tensor is proportional to the beta function of the theory. Finite theories with vanishing beta function can thus exhibit quantum scale invariance. The simplest example of this phenomenon occurs in $N = 4$ supersymmetric Yang-Mills theory. Understanding the monopole spectrum in these theories is one of the goals of these lectures. We will return later to the implications of spontaneously broken scale invariance.

For now we proceed with a discussion of the theory with non-zero potential. We want to discuss non-trivial solutions to the classical equations of motion but before doing that it will be useful to first discuss the vacuum structure of the theory. The energy density of any field configuration is given by the $(0, 0)$ component of the energy-momentum tensor,

$$\Theta_{00} = \tfrac{1}{2}\left(\mathbf{E}^a \cdot \mathbf{E}_a + \mathbf{B}^a \cdot \mathbf{B}_a + \Pi^a\Pi_a + \mathbf{D}\Phi^a \cdot \mathbf{D}\Phi_a\right) + V(\Phi) \tag{1.27}$$

[4] A canonical computation is tedious and requires the addition of improvement terms to obtain the symmetric and gauge invariant answer given here. A better procedure is to couple the theory to a background metric $g_{\mu\nu}$ and to define the energy-momentum tensor as the variation of the action with respect to the background metric.

[5] The theory is also conformally invariant but this will not play an important role in what follows, basically because spontaneous breaking of scale and conformal invariance only leads to a single Nambu-Goldstone boson.

where Π^a is the momentum conjugate to Φ, $\Pi^a = D_0\Phi^a$ and $\mathbf{E}^a$ and $\mathbf{B}^a$ are the non-abelian electric and magnetic fields,

$$\begin{aligned} E^{ai} &= F^{a0i} \\ B^{ai} &= \tfrac{1}{2}\epsilon^{ijk}F^a_{jk} \end{aligned} \tag{1.28}$$

It is clear that $\Theta_{00} \geq 0$ with equality if only if $F^{a\mu\nu} = D^\mu\Phi^a = V(\Phi) = 0$. The vacuum is thus given by a configuration with vanishing gauge field and with a constant Higgs field Φ^a with $\Phi^a\Phi_a = v^2$. [6]. A constant Higgs field breaks the gauge symmetry from $SU(2)$ down to a $U(1)$ subgroup [7]. The perturbative spectrum consists of a massless photon, massive spin one gauge bosons $W^\pm$ with mass ev and a Higgs field with a mass depending on the second derivative of the potential V at its minimum. For the previous choice of potential the mass is $m_H = \sqrt{2\lambda}v$.

We can define the Higgs vacuum to be the set of all Higgs configurations which minimize the potential,

$$M_H = \{\Phi : V(\Phi) = 0\} \tag{1.29}$$

In our example this space is just the two-sphere given by $\Phi^a\Phi_a = v^2$.

So far we have considered the vacuum configuration and the perturbative excitations about this vacuum. Now finite energy configurations need not lie in the Higgs vacuum everywhere but they must lie in M_H at spatial infinity. Thus for a finite energy configuration the Higgs field Φ^a, evaluated as $r \to \infty$, provides a map from the S^2 at spatial infinity into the S^2 of the Higgs vacuum,

$$\Phi : S^2_\infty \to M_H = S^2. \tag{1.30}$$

Such maps are characterized by an integer which measures the winding of one S^2 around the other (see Exercise 3). Mathematically, the second homotopy group of S^2 is the integers, $\pi_2(S^2) = Z$.

We have argued so far that finite energy configuration have a topological classification and that the gauge symmetry of $SU(2)$ is broken down to $U(1)$. What is the connection with magnetic monopoles? I will first give a hand-waving argument. Consider a Higgs field configuration Φ^a with winding $N \neq 0$. If the gauge field A^a_μ vanishes then we have for the total energy

$$\text{Energy} = \int d^3x \left(\tfrac{1}{2}\nabla\Phi^a \cdot \nabla\Phi_a + \tfrac{1}{2}\dot{\Phi}^a\dot{\Phi}_a + V(\Phi)\right) \geq \int d^3x\, \tfrac{1}{2}\nabla\Phi^a \cdot \nabla\Phi_a. \tag{1.31}$$

Now write the gradient term as a radial derivative plus an angular derivative

$$(\nabla\Phi^a)^2 = \left(\frac{\partial\Phi^a}{\partial r}\right)^2 + (\hat{\mathbf{r}} \times \nabla\Phi^a)^2 \tag{1.32}$$

[6] I will abuse notation by writing Φ^a for the vacuum expectation value of the operator Φ^a and hope that it will be clear from the context when I am discussing the full field and when I am discussing only its vacuum expectation value.

[7] This of course is also an abuse of terminology but one that is universal. Gauge symmetries are not true symmetries but are redundancies in our description of the configuration space of the theory. As such they are never broken.

If $N \neq 0$ then there must be non-vanishing angular derivatives of Φ^a at infinity which make the second term in (1.32) go like r^{-2} for large r. Therefore the total energy is

$$\text{Energy} > \int \frac{r^2 dr}{r^2} \tag{1.33}$$

which diverges linearly. Therefore to have finite energy configurations with $N \neq 0$ we must have non-zero gauge fields. From the above argument it is clear what we need to ensure finite energy. With non-zero gauge fields the energy involves the covariant derivative of Φ^a rather than the ordinary derivative, we can make the energy finite if there is a cancellation between the angular part of the vector potential (which must therefore fall off as $1/r$) and the angular derivative of Φ. This $1/r$ falloff in the angular component of A_μ gives rise to a non-zero magnetic field at infinity. Thus the connection between topology and magnetic charge comes about by demanding finite energy of the field configuration.

This hand-waving argument can be made somewhat more explicit. I will however not try to make it less hand-waving. The reader who wishes a more precise argument should consult the monopole reviews cited earlier. In order to have finite energy we need to ensure that the covariant derivative of Φ^a falls off faster than $1/r$ at infinity. Let us write

$$D_\mu \Phi^a = \partial_\mu \Phi^a - e\epsilon^a{}_{bc} A^b_\mu \Phi^c \sim 0 \tag{1.34}$$

to indicate that the leading $1/r$ terms must vanish at large r. Then the general solution for the gauge field, to this order, is given by

$$A^a_\mu \sim \frac{1}{ev^2}\epsilon^a{}_{bc}\Phi^b \partial_\mu \Phi^c + \frac{1}{v}\Phi^a A_\mu \tag{1.35}$$

with A_μ arbitrary.

If we now compute the leading order behavior of the non-abelian gauge field we find

$$F^{a\mu\nu} = \frac{1}{v}\Phi^a F^{\mu\nu} \tag{1.36}$$

with

$$F^{\mu\nu} = \frac{1}{ev^3}\epsilon_{abc}\Phi^a \partial^\mu \Phi^b \partial^\nu \Phi^c + \partial^\mu A^\nu - \partial^\nu A^\mu \tag{1.37}$$

and the equations of motion imply $\partial_\mu F^{\mu\nu} = \partial_\mu * F^{\mu\nu} = 0$. Thus we learn that outside the core of the monopole the non-abelian gauge field is purely in the direction of Φ^a, that is the direction of the unbroken $U(1)$. The magnetic charge of this field configuration is then

$$g = \int_{S^2_\infty} \mathbf{B}\cdot d\mathbf{S} = \frac{1}{2ev^3}\int_{S^2_\infty} dS_i\, \epsilon^{ijk}\epsilon_{abc}\Phi^a \partial_j \Phi^b \partial_k \Phi^c = \frac{4\pi N}{e} \tag{1.38}$$

with N the winding number of the Higgs field configuration.

We thus find a quantization condition

$$eg = 4\pi N. \tag{1.39}$$

This is same as the Dirac quantization condition (1.19) for even values of n in (1.19). The reason for the additional restriction on n is that in this theory we could add fields in the

fundamental **2** representation of $SU(2)$. These would carry electric charge $\pm e/2$ and the Dirac quantization condition with regard to these charges requires (1.39).

It may seem rather puzzling that we have found the same quantization method by rather different arguments. The Dirac argument just relies on what goes on outside the monopole while the quantization condition for 't Hooft Polyakov monopoles involved the topology of non-Abelian Higgs fields. The connection between these two points of view is beautiful and deep and is covered in detail in [12, 13]. Very briefly, if we are given a gauge group G broken down to a subgroup H by the Higgs field expectation value then the vacuum manifold is $M_H = G/H$ (modulo some technical assumptions) and the topology of the Higgs field at infinity is classified by $\pi_2(G/H)$. On the other hand, at infinity the gauge group is just H, and the Wu-Yang version of the Dirac monopole involves patching H gauge fields along the equator and these are classified by $\pi_1(H)$. But a famous result (proved in the references) asserts that

$$\pi_2(G/H) = \pi_1(H) \tag{1.40}$$

(as long as G is simply connected) thus providing the link between the two points of view.

1.5 Exercises for Lecture 1

E1. Consider a point electric charge e and a point magnetic charge g. Compute the field angular momentum

$$\mathbf{L} = \int d^3r\, \mathbf{r} \times (\mathbf{E} \times \mathbf{B}) \tag{1.41}$$

and

a) Show that $\mathbf{L}$ is well-defined and independent of the distance between e and g.

b) Show that demanding that the angular momentum be quantized in units of $\hbar/2$ yields the Dirac quantization condition.

E2. a) By generalizing Exercise 1 or otherwise prove the Dirac-Zwanziger-Schwinger condition for two point charges (dyons) with combined (electric,magnetic) charges (e_1, g_1) and (e_2, g_2)

$$e_1 g_2 - e_2 g_1 = 2\pi n. \tag{1.42}$$

b) Explore the allowed solutions to (1.42) assuming the existence of an electron with charges $(e, 0)$. Recall that under CP $(e, g) \to (-e, g)$. Show that there are solutions which lead to a CP violating dyon spectrum. Show that there are CP invariant solutions with dyons carrying half of the electric charge of an electron.

E3. a) If $\Phi^a \to v\hat{r}^a$ as $r \to \infty$ show that

$$N = \frac{1}{8\pi v^3} \int_{S^2_\infty} dS_i\, \epsilon^{ijk} \epsilon_{abc} \Phi^a \partial_j \Phi^b \partial_k \Phi^c = 1. \tag{1.43}$$

b) Construct a map $S^2_\infty \to S^2$ having arbitrary integer winding number N.

2 Lecture 2

2.1 Symmetric Monopoles and the Bogomol'nyi Bound

In the previous lecture we have argued that finite energy configurations with non-zero topological charge in the theory defined by (1.20) are necessarily magnetic monopoles satisfying the Dirac quantization condition. While one can argue indirectly for the existence of such solutions to the equations of motion, it would be nice to construct solutions directly. Unfortunately, in general this only turns out to be possible numerically.

To construct a solution even numerically it is necessary to make some simplifying assumptions regarding the form of the gauge and Higgs fields. We would expect the lowest energy solution to be the one of highest symmetry compatible with having non-zero topological charge. The theory defined by (1.20) is Lorentz invariant and hence rotationally invariant. Let J^i be the generators of the rotation group $SO(3)_R$. Since the scalar Higgs field must vary at infinity to have non-zero topological charge it is clear that the solution cannot be invariant under $SO(3)_R$. The Lagrangian (1.20) also is invariant under global gauge transformations by the group $SO(3)_G$ with generators T_a. Since the vacuum expectation value of the Higgs field is non-zero the monopole solution cannot be invariant under $SO(3)_G$. However, it is allowed to be invariant under the $SO(3)$ diagonal subgroup of the product of rotations and global gauge transformations $SO(3)_R \times SO(3)_G$, that is it is invariant under the generators $\mathbf{K} = \mathbf{J} + \mathbf{T}$. By imposing this $SO(3)$ symmetry as well as a Z_2 symmetry which consists of parity plus a change of sign of Φ one is left with a fairly simple ansatz in terms of two radial functions H, K:

$$\begin{aligned} \Phi^a &= \frac{\hat{r}^a}{er} H(ver) \\ A_i^a &= -\epsilon^a{}_{ij} \frac{\hat{r}^j}{er} (1 - K(ver)). \end{aligned} \tag{2.1}$$

Substituting this ansatz into the equations of motion (1.23) yields coupled differential equations for H, K which can be solved numerically subject to the boundary conditions

$$\begin{aligned} &K(ver) \to 1, \quad H(ver) \to 0, \qquad r \to 0; \\ &K(ver) \to 0, \quad H(ver)/(ver) \to 1, \qquad r \to \infty. \end{aligned} \tag{2.2}$$

In these lectures we will not need the detailed form of these solutions and for the most part will be interested in a specific limit of the equations (1.23) where an explicit solution is available. To understand the nature of this limit we first discuss a general bound on the mass of configurations with non-zero winding number known as the Bogomol'nyi bound [30].

To prove the Bogomol'nyi bound we first note that we can write the magnetic charge as

$$g = \int_{S^2_\infty} \mathbf{B} \cdot d\mathbf{S} = \frac{1}{v} \int_{S^2_\infty} \Phi^a \mathbf{B}_a \cdot d\mathbf{S} = \frac{1}{v} \int \mathbf{B}^a \cdot (\mathbf{D}\Phi)_a \, d^3r \tag{2.3}$$

using the Bianchi identity $\mathbf{D} \cdot \mathbf{B}^a = 0$ and integration by parts. Then if we consider a static configuration with vanishing electric field the energy (mass) of the configuration is given

by

$$\begin{aligned} M_M &= \int d^3r \left(\tfrac{1}{2}(\mathbf{B}^a \cdot \mathbf{B}_a + \mathbf{D}\Phi^a \cdot \mathbf{D}\Phi_a) + V(\Phi)\right) \geq \int d^3r \, \tfrac{1}{2}(\mathbf{B}^a \cdot \mathbf{B}_a + \mathbf{D}\Phi^a \cdot \mathbf{D}\Phi_a) \\ &= \tfrac{1}{2} \int d^3r \, (\mathbf{B}^a - \mathbf{D}\Phi^a) \cdot (\mathbf{B}_a - \mathbf{D}\Phi_a) + vg \end{aligned} \tag{2.4}$$

using (2.3). We thus have the bound

$$M_M \geq vg \tag{2.5}$$

with equality iff $V(\Phi) \equiv 0$ and the first-order Bogomol'nyi equation

$$\mathbf{B}^a = \mathbf{D}\Phi^a \tag{2.6}$$

is satisfied. Note that the bound has been derived for positive magnetic charge. For negative magnetic charge we get (2.5) with a minus sign on the right. Thus the general bound is $M_M \geq |vg|$. In Exercise 4 this bound is generalized to include configurations with non-zero electric field as well.

To saturate the bound (2.5) we require that the potential vanish identically and that the Bogomol'nyi equation (2.6) be satisfied. Let us first discuss vanishing potential. Classically we are free to choose $V(\Phi) = 0$ but we know that quantum mechanically there will be corrections to V [31]. Eventually we will consider supersymmetric theories which have potentials with exact flat directions protected by supersymmetry. For the meantime we will consider just the classical theory and impose $V(\Phi) \equiv 0$ by hand. The next question is whether symmetry breaking makes sense with $V(\Phi) \equiv 0$. We can impose as a boundary condition that $\Phi^a\Phi_a \to v^2$ as $r \to \infty$ for arbitrary v. Although there is no potential, a change of the theory from one value of v to another value requires changing the Higgs field at infinity. Since we are in infinite volume such a motion requires infinite action, even in the absence of a potential. Therefore for each value of v the imposition of this boundary condition at infinity gives a well defined Hilbert Space which does not mix with Hilbert spaces built on other values of v. In other words each value of v determines a superselection sector of the theory.

2.2 The Prasad-Sommerfield Solution

Following the previous discussion we now proceed to look for a solution of (2.6) with spherical symmetry. The ansatz (2.1) when substituted into (2.6) yields the equations

$$yK' = -KH; \qquad yH' = H - (K^2 - 1) \tag{2.7}$$

with $y = ver$ and $H' = dH/dy$. Manipulation of these equations yields the solution [32]

$$\begin{aligned} H(y) &= y \coth y - 1 \\ K(y) &= \frac{y}{\sinh y}. \end{aligned} \tag{2.8}$$

The long range behavior of this solution is important. At large r, K vanishes exponentially at distances greater than $1/(ev) = 1/M_W$ with M_W the mass of the $W^\pm$ gauge

bosons resulting from spontaneous symmetry breaking. Physically this means that there are $W^{\pm}$ fields excited in the core of the monopole, but that outside the core the magnetic field falls like $1/r^2$ as required for a magnetic monopole. The form of the Higgs field is also interesting. There is an exponentially decaying piece, but also a piece which falls of only as $1/r$. For large r we have

$$\Phi^a \to v\hat{r}^a - \frac{\hat{r}^a}{er}. \tag{2.9}$$

This power law falloff is due to the massless dilaton field in this scale invariant limit. To define the dilaton field $\mathcal{D}$ we write fluctuations of Φ^a about the asymptotic monopole configuration in the form

$$\Phi^a = v\hat{r}^a e^{\mathcal{D}} = v\hat{r}^a + v\hat{r}^a\mathcal{D} + \cdots \tag{2.10}$$

We can then define a dimensionless "dilaton charge" as

$$Q_{dil} = v\int_{S^2_\infty} \nabla\mathcal{D}\cdot d\mathbf{S} \tag{2.11}$$

and using (2.9) we see that for the monopole solution $Q_{dil} = 4\pi/e = g = M_M/v$.

2.3 Collective Coordinates and the Monopole Moduli Space

Given a classical solution in field theory one often finds that the solution is part of a multi-parameter family of solutions with the same energy. The parameters labeling the different degenerate solutions are called collective coordinates or moduli and the space of solutions of fixed energy (and topological charge) is called the moduli space of solutions. Before discussing the general situation it will be useful to identify the four collective coordinates of a charge one BPS monopole [33].

To start with, in (2.8) we have constructed a monopole sitting at the origin. By translation invariance of (1.20) a monopole sitting at any other point in R^3 is also a solution with the same energy. If we let $\mathbf{X}$ denote this center of mass collective coordinate then the general solution is

$$\Phi^a_{\rm cl}(\mathbf{r}+\mathbf{X}), \qquad A^a_{i\,\rm cl}(\mathbf{r}+\mathbf{X}) \tag{2.12}$$

with the classical solutions at $\mathbf{X} = 0$ given by (2.1). We can construct a slowly moving monopole by letting $\mathbf{X}$ depend on time so that we have fields $A^a_{i\,\rm cl}(\mathbf{r}+\mathbf{X}(t))$. This time dependence will of course give rise to an electric field and the energy of the monopole will exceed the Bogomol'nyi bound by the kinetic energy of the monopole. This is precisely what should happen for motion in the moduli space, the potential terms stay constant and the kinetic terms are proportional to the velocity of the motion along the moduli space.

The remaining collective coordinate is somewhat more subtle. At this point it is useful to recall some basic facts about the configuration space of gauge theories. In gauge theory it is important to make a distinction between *small* gauge transformations g which are those approaching the identity at spatial infinity and *large* gauge transformations which do not approach the identity at spatial infinity. These play different roles in gauge theory. In

particular given the space of gauge and Higgs fields $\mathcal{A} = (A, \Phi)$ the physical configuration space is given by

$$\mathcal{C} = \mathcal{A}/\mathcal{G} \tag{2.13}$$

where $\mathcal{G}$ is the group of small gauge transformations. Thus physical states are invariant under small gauge transformations and they do not act as symmetries of $\mathcal{C}$. Rather, they describe a redundancy in our description of the theory when we work just in $\mathcal{A}$. Large gauge transformations on the other hand do not identify points in $\mathcal{C}$ but instead act as true symmetries which relate different points in $\mathcal{C}$ with the same properties.

With this in mind we will try to identify an additional collective coordinate associated to global $U(1)$ electromagnetic gauge transformations. Heuristically we expect such a collective coordinate because the monopole solution contains excitations of the electrically charge $W^{\pm}$ fields in its core.

We will work in $A_0 = 0$ gauge with a BPS monopole configuration A_i, Φ obeying the equation $B_i = D_i\Phi$. A deformation of this solution $\delta A_i(\mathbf{x}, t), \delta\Phi(\mathbf{x}, t)$ which keeps the potential energy fixed must obey the linearized Bogomol'nyi equation

$$\epsilon_i{}^{jk} D_j \delta A_k = D_i \delta\Phi - [\delta A_i, \Phi] \tag{2.14}$$

and the Gauss law constraint

$$D^i \delta\dot{A}_i - [\Phi, \delta\dot{\Phi}] = 0 \tag{2.15}$$

The unique solution (modulo small gauge transformations) is

$$\begin{aligned} \delta A_i &= D_i(\chi(t)\Phi) \\ \delta\Phi &= 0 \\ \delta A_0 &= D_0(\chi(t)\Phi) - \dot{\chi}\Phi \end{aligned} \tag{2.16}$$

where $\chi(t)$ is an arbitrary function of time. Note that A_0 vanishes identically, it has been written in the form (2.16) to make clear the relation with gauge transformations. This solution has the following properties

1. It obeys (2.14) and (2.15)

2. For $\dot{\chi} = 0$ the deformation is by a large gauge transformation with $g = e^{\chi\Phi}$ so that χ is a physical zero mode.

3. For $\dot{\chi} \neq 0$ the linearized Bogomol'nyi equation is still satisfied so there is no change in the potential energy $(\mathbf{B}^2 + (\mathbf{D}\Phi)^2)$ but there is an increase in the kinetic energy $(\mathbf{E}^2)$. If we think of the configuration space as a mountain range then the moduli space is a flat valley. Motions that stay purely along the valley have fixed potential energy but variable kinetic energy, as we have found.

4. Since the unbroken gauge group $U(1)$ is compact, χ is a periodic coordinate. Therefore the one monopole moduli space is topologically $\mathcal{M}_1 = R^3 \times S^1$.

So far we have limited our discussion to monopoles with $N = 1$. It is at first sight not clear whether we expect static solutions to exist with $N \geq 1$ and if they do what the collective coordinates should be. Physically, we can argue as follows. Away from the BPS limit the photon is the only massless field in the theory. Multi-monopoles of the same sign magnetic charge which are well separated will thus experience a Coulomb repulsion and we thus do not expect static solutions for such a configuration. On the other hand in the BPS limit the Higgs field is really a dilaton of spontaneously broken scale invariance (at least classically) and we have seen that the one monopole solution carries a charge under this Higgs field. Since Higgs exchange is always attractive, there can be a cancellation between the Coulomb repulsion and Higgs attraction. Classically this cancellation does occur as a consequence of the fact that the magnetic charge and dilaton charge of the monopole are equal as was found below (2.11). This equality is not accidental but has its roots in spontaneously broken scale invariance which forces the dilaton charge of any state to equal its mass.

Thus we might expect on physical grounds that there are solutions given by well separated static monopoles and that for magnetic charge k the moduli space is $4k$ dimensional with the collective coordinates being the locations of the k monopoles and their dyon degrees of freedom. This is correct, at least for large separation, although the above hardly constitutes a serious argument. A careful analysis of the issue would take us to far afield. It basically involves the use of index theory to count perturbations of the Bogomol'nyi equations. A discussion suitable for physicists may be found in [34], a mathematical proof is given in [35].

Suitably explicit multimonopole solutions are hard to come by but in spite of this it is possible to say some general things about the structure of the multi-monopole moduli space. In the final lecture we will discuss the structure of the two-monopole moduli space in some detail.

Before proceeding it is useful to make use of a connection between the Bogomol'nyi equations on R^3 and the self-dual Yang-Mills equations on R^4. If we write $A_4 = \Phi$ then we can rewrite the Bogomol'nyi equation (2.6) as

$$F_{ab} = *F_{ab} \tag{2.17}$$

where $a, b = 1 \cdots 4$, we work on R^4 with coordinates x_1, x_2, x_3, x_4 and Euclidean signature so that $** = 1$, and we restrict ourselves to configurations which are independent of x_4. This suggest that there is a deep connection between the problem of solving the Bogomol'nyi equations and the problem of solving the self-dual Yang-Mills equations. We will just use this connection to simplify the notation. For example Gauss' Law reads

$$D^a \dot{A}_a = 0 \tag{2.18}$$

and gauge transformations take the form

$$\delta A_a = D_a \Lambda \tag{2.19}$$

where it is always understood that all quantities are independent of x_4.

Now the k monopole moduli space $\mathcal{M}_k$ is defined as the space of solutions to the Bogomol'nyi equations having topological charge k. Tangent vectors to $\mathcal{M}_k$, $\delta_\alpha A_a$, are deformations of a given charge k solution $A_a \to A_a + \delta_\alpha A_a$ which satisfy the linearized Bogomol'nyi equations

$$D_a \delta_\alpha A_b - D_b \delta_\alpha A_a = \tfrac{1}{2}\epsilon_{ab}{}^{cd}(D_c \delta_\alpha A_d - D_d \delta_\alpha A_c) \tag{2.20}$$

and are orthogonal to (small) gauge transformations

$$D^a \delta_\alpha A_a = 0 \tag{2.21}$$

so that they leave one in the physical configuration space.

Given such a tangent vector the metric on $\mathcal{M}_k$ is

$$\mathcal{G}_{\alpha\beta} = -\int d^3x \, \mathrm{Tr}\, \delta_\alpha A_a \, \delta_\beta A^a \tag{2.22}$$

This metric is inherited from the action for the underlying gauge theory. To see this, imagine we are given a charge k BPS monopole solution $A_a(\mathbf{x}, z^\alpha)$ depending on $4k$ collective coordinates z^α. By definition the potential energy is independent of the z^α. Now one might think that we could construct tangent vectors to $\mathcal{M}_k$ simply by differentiating with respect to the z^α. This is not quite correct because there is no guarantee that the resulting change to A_a is orthogonal to gauge transformations, in other words differentiating with respect to the z^α may include a gauge transformation. However we can always cure this by undoing the gauge part of this variation by writing the tangent vector as

$$\delta_\alpha A_a = \frac{\partial A_a}{\partial z^\alpha} - D_a \epsilon_\alpha \tag{2.23}$$

where $\epsilon_\alpha(\mathbf{x}, z^\beta)$ is a gauge parameter chosen to ensure that (2.21) is satisfied.

To construct the metric we consider slow time dependent variations of the collective coordinates $z^\alpha(t)$. If we write

$$A_a(\mathbf{x}, z^\alpha(t)), \quad A_0 = \dot{z}^\alpha \epsilon_\alpha \tag{2.24}$$

then $F_{0a} = \dot{z}^\alpha \delta_\alpha A_a$ and the action is

$$S = -\frac{1}{2}\int d^3x \, dt \, \mathrm{Tr}\, F_{0a} F_0{}^a = \frac{1}{2}\int dt \, \mathcal{G}_{\alpha\beta}\dot{z}^\alpha \dot{z}^\beta \tag{2.25}$$

with $\mathcal{G}_{\alpha\beta}$ as in (2.22).

2.4 Exercises for Lecture 2

E4. For a dyon with electric and magnetic charge (q, g) prove the bound

$$M \geq v(q^2 + g^2)^{1/2}. \tag{2.26}$$

E5. Following the discussion in the lecture derive the action for the collective coordinates $\mathbf{X}, \chi$ of the charge one BPS monopole. Quantize this action to deduce the spectrum of states in the magnetic charge one sector. Note that the states you obtain by quantizing the dyon collective coordinate χ consist of an infinite tower of states of increasing mass and electric charge. Thus in the monopole sector electric charge is classically continous, but is quantized when treated quantum mechanically.

E6. Compute the generators J^i and T_a of rotations and global gauge transformations and verify that the ansatz (2.1) is left invariant by the action of $\mathbf{J} + \mathbf{T}$.

E7. It is possible to view the gauge parameter $\epsilon_\alpha(x, z)$ as a connection on $\mathcal{M}_k$ with covariant derivative $s_\alpha = \partial_\alpha - [\epsilon_\alpha, \quad]$. Show that $\delta_\alpha A_a$ can then be viewed as a mixed component of the curvature of the connection (A_a, ϵ_α) on $R^4 \times \mathcal{M}_k$.

E8. Compute the dilaton charge of a massive W^+ boson at rest at the origin and show that it is equal to m_W/v. Show that this follows from the general theory of spontaneously broken scale invariance.

3 Lecture 3

3.1 Witten Effect

There is a famous term, the θ term, which can be added to the Lagrangian for Yang-Mills theory without spoiling renormalizability. It is given by

$$\mathcal{L}_\theta = -\frac{\theta e^2}{32\pi^2} F^a_{\mu\nu} * F^{\mu\nu}_a. \tag{3.1}$$

This interaction violates P and CP but not C. Since it preserves C we may expect that it is consistent with the existence of a duality symmetry of the theory. As is well known [36], this term is a surface term and does not affect the classical equations of motion. There is however θ dependence in instanton effects which involve non-trivial long-range behavior of the gauge fields. As was realized by Witten [37], in the presence of magnetic monopoles θ also has a non-trivial effect, it shifts the allowed values of electric charge in the monopole sector of the theory.

I will give two explanations of this effect, the first is borrowed from Coleman [14], the second from Witten [37]. First consider pure electromagnetism. Then the θ term reduces to the QED interaction

$$\mathcal{L}_\theta = \frac{\theta e^2}{8\pi^2} \mathbf{E} \cdot \mathbf{B}. \tag{3.2}$$

Now consider this interaction in the presence of a (Dirac) magnetic monopole. Writing the fields as a monopole field plus corrections we have

$$\begin{aligned} \mathbf{E} &= \nabla A_0 \\ \mathbf{B} &= \nabla \times \mathbf{A} + \frac{g}{4\pi} \frac{\hat{\mathbf{r}}}{r^2}. \end{aligned} \tag{3.3}$$

Substituting into the action density (3.2) we obtain

$$\begin{aligned} L_\theta &= \int d^3r \mathcal{L}_\theta = \frac{\theta e^2}{8\pi^2} \int d^3r \, \nabla A_0 \cdot \left(\nabla \times \mathbf{A} + \frac{g}{4\pi} \frac{\hat{\mathbf{r}}}{r^2} \right) \\ &= -\frac{\theta e^2 g}{32\pi^3} \int d^3r \, A_0 \nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} = -\frac{\theta e^2 g}{8\pi^2} \int d^3r \, A_0 \delta^3(\mathbf{r}) \end{aligned} \tag{3.4}$$

which we recognize as the coupling of the scalar potential A_0 to an electric charge of magnitude $-\theta e^2 g/8\pi^2$ located at the origin. In other words, the magnetic monopole has acquired an electric charge. For a minimal charge monopole with $eg = 4\pi$ the electric charge of the monopole is $-e\theta/2\pi$. Although this derivation gives the correct answer, one may feel a bit uneasy about the method used. We don't really know what is going on at the origin for a Dirac monopole yet this calculation suggests a delta function electric charge density located at the origin.

A more fundamental derivation which applies to the full $SU(2)$ gauge theory and which does not suffer from this ambiguity runs as follows. We have seen that the dyon collective coordinate of the monopole allows it to carry electric charge. The dyon collective coordinate arises through $U(1)$ gauge transformations which are constant at infinity. We now consider these transformations in the presence of a theta term. We are interested in gauge transformations, constant at infinity, which are rotations in the $U(1)$ subgroup of $SU(2)$ picked out by the gauge field. That is, rotations in $SU(2)$ about the axis $\hat{\Phi}^a = \Phi^a/|\Phi|$. The action of such an infinitesimal gauge transformation on the field is

$$\delta A^a_\mu = -\frac{1}{ev}(D_\mu \Phi)^a \tag{3.5}$$

with Φ the background monopole Higgs field. Let $\mathcal{N}$ denote the generator of this gauge transformation. Then if we rotate by 2π about the $\hat{\Phi}$ axis we must get the identity [8]. That is, physical states must obey

$$e^{2\pi i \mathcal{N}} = 1. \tag{3.6}$$

It is straightforward to compute $\mathcal{N}$ using the Noether method,

$$\mathcal{N} = \frac{\partial \mathcal{L}}{\partial(\partial_0 A^a_\mu)} \delta A^a_\mu \tag{3.7}$$

with δA^a_μ given by (3.5). Including the theta term one finds

$$\mathcal{N} = \frac{Q}{e} + \frac{\theta e g}{8\pi^2} \tag{3.8}$$

where

$$\begin{aligned} g &= \frac{1}{v} \int d^3x \, D_i \Phi^a B^i_a \\ Q &= \frac{1}{v} \int d^3x \, D_i \Phi^a E^i_a \end{aligned} \tag{3.9}$$

[8] At this point we are working in a theory with gauge group $SU(2)/Z_2 = SO(3)$ since all states are in the adjoint representation. In this theory a 2π rotation gives the identity. Later, when we consider $SU(2)$ and states in the fundamental representation this condition will be modified since then a 2π rotation gives an element of the center of $SU(2)$ which acts non-trivially on the fundamental representation.

are the magnetic and electric charge operators respectively. The condition (3.6) thus implies that

$$Q = n_e e - \frac{e\theta n_m}{2\pi} \tag{3.10}$$

where n_e is an arbitrary integer and $n_m = eg/4\pi$ determines the magnetic charge of the monopole.

3.2 Montonen-Olive and SL(2,Z) Duality

Let us pause for a moment to see what we have accomplished in trying to establish a duality between electric and magnetic degrees of freedom. In the BPS limit at $\theta = 0$ we have a classical spectrum indicated in the table below

Table 1

	Mass	(Q_e, Q_m)	Spin
Higgs	0	$(0,0)$	0
Photon	0	$(0,0)$	1
$W^{\pm}$	ve	$(e,0)$	1
$M^{\pm}$	vg	$(0,g)$	0

As is evident from the table, all of these states saturate the Bogomol'nyi bound $M \geq v\sqrt{Q_e^2 + Q_m^2}$ with $Q_m = 4\pi n_m/e$ and $Q_e = n_e e - e n_m \theta/2\pi$ (with $\theta = 0$ for the moment). At weak coupling, where this analysis should be a good first order approximation to the full quantum answers, we have

$$M_W = ev \ll v, \quad M_M = gv = \frac{4\pi}{e} v \gg v \tag{3.11}$$

so although we have constructed a theory with both electric and magnetic charges, monopoles are much heavier than W bosons at weak coupling. However we would expect that to get a dual theory we would also have to exchange the role of electric and magnetic charge. Given the quantization condition this implies that we should look for a duality transformation which acts on the fields as in (1.8) but also takes

$$e \to g \equiv \frac{4\pi}{e} \tag{3.12}$$

and relabels electric and magnetic states.

Based on the classical spectrum shown in Table 1 and some other arguments Montonen and Olive proposed that this should be an exact duality of the $SO(3)$ Yang-Mills-Higgs theory in the BPS limit [1]. However, as noted by the authors of [1], there are some obvious problems with this proposal. They are:

1. Quantum corrections would be expected to generate a non-zero potential $V(\Phi)$ even if one is absent classically and should also modify the classical mass formula. Thus there is no reason to think that the duality of the spectrum should be maintained by quantum corrections.

2. The W bosons have spin one while the monopoles are rotationally invariant indicating that they have spin zero. Thus even if the mass spectrum is invariant under duality, there will not be an exact matching of states and quantum numbers.

3. The proposed duality symmetry seems impossible to test since rather than acting as a symmetry of a single theory it relates two different theories, one of which is necessarily at strong coupling where we have little control of the theory.

As we will see later, the first two problems are resolved by embedding the theory into $N = 4$ super Yang-Mills theory [38]. The third problem is still with us in that there are few concrete ways to test the proposal. However there *are* non-trivial tests and the first of these arose by first considering an extension of duality to a larger set of transformations.

It is not hard to see that if the basic duality idea is correct then it should have an interesting extension when the effects of a non-zero theta angle are included. Including the theta term, the Lagrangian we are considering is determined by two real parameters, e and θ. We can write the Lagrangian as

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \frac{\theta e^2}{32\pi^2}F^{\mu\nu} * F_{\mu\nu} - \frac{1}{2}D^\mu\Phi D_\mu\Phi \\ &\equiv -\frac{1}{32\pi}\,\mathrm{Im}(\frac{\theta}{2\pi} + \frac{4\pi i}{e^2})(F^{\mu\nu} + i * F^{\mu\nu})(F_{\mu\nu} + i * F_{\mu\nu}) - \frac{1}{2}D^\mu\Phi D_\mu\Phi\end{aligned} \tag{3.13}$$

We thus see that the Lagrangian can be written in terms of a single complex parameter

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{e^2}. \tag{3.14}$$

As an aside, note that n instanton effects in this theory are weighted by $e^{2\pi i n\tau}$.

Since physics is periodic in θ with period 2π the transformation

$$\tau \to \tau + 1 \tag{3.15}$$

should leave physics invariant up to a relabeling of states. At $\theta = 0$ the duality transformation (3.12) is given in terms of τ by

$$\tau \to -\frac{1}{\tau}. \tag{3.16}$$

It thus seems reasonable to suspect that at arbitrary θ the full duality group is generated by transformations of the form (3.15) and (3.16). It is a well known fact that these two transformations generate the group $SL(2, Z)$ of projective transformations

$$\tau \to \frac{a\tau + b}{c\tau + d}, \quad a, b, c, d \in Z, \quad ad - bc = 1. \tag{3.17}$$

Note that since $e^2 \geq 0$, τ naturally lives on the upper half plane $\mathrm{Im}\,\tau \geq 0$. Furthermore, one can check that one can use $SL(2, Z)$ transformations to map any τ in the upper half plane into the fundamental region defined by $-1/2 \leq \mathrm{Re}\tau \leq 1/2$ and $|\tau| > 1$.

In order for (3.17) to be a symmetry we know that there must in addition be a relabeling of states. From (3.10) we see that the transformation (3.15) shifts the electric charge by

-1 (for $n_m - 1$) and we know from the earlier discussion that the transformation (3.16) requires an exchange of electric and magnetic quantum numbers. Putting these two facts together we deduce that the action of $SL(2,Z)$ on the quantum numbers should be

$$\begin{pmatrix} n_e \\ n_m \end{pmatrix} \to \begin{pmatrix} a & -b \\ c & -d \end{pmatrix} \begin{pmatrix} n_e \\ n_m \end{pmatrix} \tag{3.18}$$

Finally, let us consider the spectrum of states saturating the BPS bound $M^2 \geq v^2(Q_e^2 + Q_m^2)$. We know the allowed values of Q_e and Q_m are

$$\begin{aligned} Q_m &= \frac{4\pi}{e} n_m \\ Q_e &= n_e e - n_m \frac{e\theta}{2\pi}. \end{aligned} \tag{3.19}$$

Substituting these into the formula for M^2 and writing the result in terms of τ yields

$$M^2 \geq 4\pi v^2 \begin{pmatrix} n_e, & n_m \end{pmatrix} \frac{1}{\operatorname{Im}\tau} \begin{pmatrix} 1 & -\operatorname{Re}\tau \\ -\operatorname{Re}\tau & |\tau|^2 \end{pmatrix} \begin{pmatrix} n_e \\ n_m \end{pmatrix} \tag{3.20}$$

It is left as an exercise to verify that the mass formula in this form is invariant under $SL(2,Z)$ transformations.

The extension of electromagnetic duality to $SL(2,Z)$ that we have uncovered is usually referred to as S-duality. The name is a historical accident. Although this extension was first discovered in the context of lattice models [39] it was discussed as a symmetry of $N = 4$ Yang-Mills theory first in the low-energy limit of toroidal compactifications of string theory [40]. In that context the variable τ becomes a dynamical field usually denoted by S and the $SL(2,Z)$ transformations of S were called S-duality to distinguish it from other $SL(2,Z)$ transformations in string theory which are (superficially) unrelated.

3.3 Exercises for Lecture 3

E9. Carry out the computation of the generator $\mathcal{N}$ using the Noether method. Verify that dyons with electric charge $Q = ne - e\theta/2\pi$ satisfy the DSZ quantization condition but violate CP. Show that the interaction $\mathcal{L}_\theta$ violates CP.

E10. Find two points on the boundary of the fundamental region described in the text which are left fixed by some element of $SL(2,Z)$ other than the identity. What order are these elements of $SL(2,Z)$ and what are the values of θ and e^2 at the two fixed points?

E11. Show that M^2 is left invariant by the $SL(2,Z)$ transformation given by (3.17) and (3.18). For those familiar with string theory note the close connection between the form of M^2 and the Poincare metric on the upper half plane.

E12. I have been sloppy about the precise group which is acting in (3.17) and (3.18). Show that there are order two elements which act trivially on the couplings but non-trivially on the charges as in (3.18). What do these order two elements correspond to physically?

4 Lecture 4

4.1 Monopoles and fermions

As we have seen in the previous discussion, duality is inherently a quantum symmetry since it relates weak coupling to strong coupling. As such we cannot hope to understand it easily unless we are working in a theory where quantum effects are under rather precise control. At our current level of understanding this limits us to theories with supersymmetry, and the more supersymmetry, the more control we have of the dynamics. Supersymmetry involves the addition of fermion fields with special couplings. However many of the features of fermions in monopole backgrounds are independent of supersymmetry. Thus we will start out with a general discussion of the effects of fermions and then later generalize our results to the supersymmetric context.

We will first consider Dirac fermions with couplings to the fields appearing in (1.20) determined by the Lagrangian

$$\mathcal{L}_\psi = -\bar{\psi}_n \gamma^\mu (D_\mu \psi)^n - i\bar{\psi}_n (T_a)^n{}_m \Phi^a \psi^m \tag{4.1}$$

with $(T_a)^n{}_m$ the anti-Hermitian generators of $SU(2)$ in the representation $\mathbf{r}$. We will consider only fundamental and adjoint fermions in which case we take

$$(T_a)^n{}_m = -\frac{i}{2}(\tau_a)^n{}_m \quad n, m = 1, 2 \tag{4.2}$$

with τ_a the Pauli matrices or

$$(T_a)^n{}_m = \epsilon^n{}_{am} \quad n, m = 1, 2, 3. \tag{4.3}$$

It will also be convenient following [41] to use a representation of the gamma matrices with

$$\gamma^0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \qquad \gamma^i = \begin{pmatrix} -\sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix} \tag{4.4}$$

obeying $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, with $\bar{\psi} = i\psi^\dagger \gamma^0$. We will also write $\alpha^i = -\gamma^0 \gamma^i$.

The Dirac equation is then

$$(\gamma^\mu D_\mu + \Phi)\psi = 0. \tag{4.5}$$

For a monopole configuration with $A_0 = 0$ we can look for stationary solutions of the form $\psi(\mathbf{x}, t) = e^{iEt}\psi(\mathbf{x})$. Writing ψ in terms of two-component spinors

$$\psi = \begin{pmatrix} \chi^+ \\ \chi^- \end{pmatrix} \tag{4.6}$$

we then have the coupled equations

$$\begin{aligned} \not{D}\chi^- \equiv (i\sigma^i D_i + \Phi)\chi^- &= E\chi^+ \\ \not{D}^\dagger \chi^+ \equiv (i\sigma^i D_i - \Phi)\chi^+ &= E\chi^- \end{aligned} \tag{4.7}$$

Now (4.7) will have solutions with $|E| > 0$ and may also have solutions with $E = 0$. We can quantize the fermion fluctuations about a magnetic monopole by expanding ψ in terms of eigenfunctions of the Dirac operator (4.7) and then interpreting the coefficients multiplying the eigenfunctions as creation and annihilation operators with anti-commutation relations which follow from the canonical anti-commutation relations of ψ. Modes with $|E| > 0$ will thereby lead to configurations with energy greater than the ground state energy of the monopole. On the other hand, if (4.7) has solutions with $E = 0$ then the states created by the corresponding creation operators will be degenerate in energy with the original monopole solution. Thus we can view the zero energy eigenfunctions of the Dirac operator as "fermionic collective coordinates" in the sense that they describe Grassmann valued deformations of the monopole which keep the energy fixed.

Thus to study the structure of the monopole ground state we must study the zero energy solutions of (4.7), that is we want the solutions of $\not{D}\chi^+ = 0$ and $\not{D}^\dagger\chi^- = 0$, the kernels of $\not{D}$ and $\not{D}^\dagger$. It is easy to see that the $\ker \not{D}^\dagger = \{0\}$ using the fact that $\ker \not{D}^\dagger \subset \ker \not{D}\not{D}^\dagger$ and that $\not{D}\not{D}^\dagger$ is a positive definite operator. On the other hand $\ker \not{D}$ is non-zero in a monopole background and can be computed using an index theorem of Callias [42] which gives

$$\dim \ker \not{D} - \dim \ker \not{D}^\dagger = A(r) n_m \tag{4.8}$$

with n_m the winding number of the Higgs field (the monopole charge) and $A(r)$ a constant depending on the representation of the fermion fields and the ratio of the magnitude of a bare fermion mass to the Higgs expectation value. In the examples we are discussing $A = 1$ for fundamental fermions and $A = 2$ for adjoint fermions.

While both fundamental and adjoint fermions have zero modes in a monopole background, their consequences are somewhat different so we discuss the two cases separately.

4.2 Monopoles coupled to isospinor fermions

For fundamental fermions in the **2** of $SU(2)$ a charge one monopole has a single fermion zero mode according to (4.8). To be precise, there is a single zero mode wave function but since the fermion ψ does not obey any reality condition, the coefficient multiplying the zero mode should be taken complex. We thus have the expansion

$$\psi = a_0 \psi_0 + \text{non-zero modes} \tag{4.9}$$

and the anti-commutation relations for ψ imply

$$\{a_0^\dagger, a_0\} = 1, \quad \{a_0, a_0\} = \{a_0^\dagger, a_0^\dagger\} = 0. \tag{4.10}$$

To construct the monopole ground state we start with a ground state $|\Omega\rangle$ with $a_0|\Omega\rangle = 0$ and then act with $a_0^\dagger$. This gives a two-fold degenerate ground state consisting of the two states.

$$|\Omega\rangle, \qquad a_0^\dagger|\Omega\rangle \tag{4.11}$$

Given this degeneracy it is natural to ask whether there are quantum numbers which distinguish the two ground states. The standard answer, given in [41], is that this theory

has a fermion number conjugation symmetry

$$\psi^n \to \begin{pmatrix} \sigma^2 & 0 \\ 0 & -\sigma^2 \end{pmatrix} (\tau_2)^n{}_m \psi^{*m} \tag{4.12}$$

which changes the sign of the $U(1)$ fermion number charge of any state. The two degenerate monopole states differ by one unit of fermion number since one obtains one from the other by acting with fermion creation operators which carry fermion number one. On the other hand the fermion number conjugation symmetry can only be respected if the two states carry opposite fermion number. Thus it is argued in [41] that the two states in (4.11) have fermion number $\pm 1/2$. However this argument suffers from the following difficulty. The discrete symmetry (4.12) is a classical symmetry of the theory which forbids a Dirac mass term for the matter fermions. On the other hand one can show that instanton effects will generate such a Dirac mass term which is related to the fact that the symmetry (4.12) involves a discrete chiral transformation. In other words, the fermion number conjugation symmetry is anomalous and is not an exact quantum symmetry of the full theory. However, at $\theta = 0$ the theory is also CP invariant. CP takes the magnetic charge into itself but changes the sign of the fermion number charge. Thus CP invariance enforces the assignment of fermion number $\pm 1/2$ to the two ground states.

A more interesting example of charge fractionalization occurs if we take N_f flavors of Dirac fermion coupled to a monopole background with Lagrangian

$$\mathcal{L}_\psi = \sum_{I=1}^{N_f} \left(-\bar{\psi}_I \gamma^\mu D_\mu \psi^I - if \bar{\psi}_I \Phi \psi^I \right) \tag{4.13}$$

In this theory the $U(1) \sim O(2)$ fermion number symmetry is extended to a $O(2N_f)$ symmetry. Of this symmetry only a $SU(N_f) \times U(1)$ symmetry is manifest in the Lagrangian (4.13). To see the full $O(2N_f)$ symmetry note that we can write the Dirac fermions ψ^I in terms of $2N_F$ Weyl fermions $\chi^a, a = 1 \ldots 2N_f$. Furthermore, since the doublet representation of $SU(2)$ is pseudoreal we can take the χ^a to all be say left-handed under the Lorentz group and to all transform the same way under the $SU(2)$ gauge symmetry. If we write (4.13) in this basis then the quadratic terms involving the χ^a are $O(2N_f)$ invariant and $O(2N_f)$ commutes with both the Lorentz group and the gauge group.

Thus we have $2N_f$ Weyl fermions transforming as a vector of $O(2N_f)$ and in the zero mode expansion of ψ^I we will have

$$\psi^I = a_0^I \psi_0 + \text{non-zero modes} \tag{4.14}$$

To see what the consequences are for the spectrum it is useful to first rephrase the results we found for $N_f = 1$. There we could trade the operators $(a_0, a_0^\dagger)$ for a pair of self-conjugate operators (b_0^1, b_0^2) by writing

$$\begin{aligned} a_0 &= \frac{1}{\sqrt{2}}(b_0^1 + i b_0^2) \\ a_0^\dagger &= \frac{1}{\sqrt{2}}(b_0^1 - i b_0^2) \end{aligned} \tag{4.15}$$

The b_0^i, $i = 1, 2$ then obey the Clifford algebra

$$\{b_0^i, b_0^j\} = \delta^{ij}. \tag{4.16}$$

The ground state must furnish a representation of this Clifford algebra and since the smallest representation is two-dimensional we again conclude that the ground state is two-fold degenerate.

We can apply the same technique for arbitrary N_f in which case we end up with operators b_0^a, $a = 1 \cdots 2N_f$ obeying

$$\{b_0^a, b_0^b\} = \delta^{ab}. \tag{4.17}$$

Representations of this Clifford algebra have dimension $2^{2N_f/2} = 2^{N_f}$, that is the monopole ground state is now a spinor of $SO(2N_f)$! This is precisely the phenomenon that allows one to construct spacetime fermions in the Ramond sector of superstring theory.

Since we have added fermions and changed the global structure of the gauge group (from $SO(3)$ to $SU(2)$) we should also go back and reanalyze the constraint (3.6) which followed from the action of global $U(1)$ charge rotations. We can repeat most of the previous discussion but with one change, since we are now working in $SU(2)$ and not $SO(3)$ a rotation by 2π about some axis does not give the identity but rather gives the non-trivial element of the center of $SU(2)$ which acts on spinor representations as -1. If we denote this operator by $(-1)^H$ following [5] then we have the relation

$$\exp\left(2\pi i\left(\frac{Q}{e} + \frac{\theta n_m}{2\pi}\right)\right) = (-1)^H \tag{4.18}$$

If $Q = n_e e - e n_m \theta/2\pi$ then (4.18) says that there is a correlation between the action of the center of $SU(2)$ and the electric charge, states in spinor representations of $SU(2)$ have n_e half an odd integer and states transforming trivially under the center must have n_e integer. This of course agrees with our expectations in the zero magnetic charge sector of the theory. In the monopole sector the implications of (4.18) are as follows. Since $(-1)^H$ acts as the center of $SU(2)$ and the fermion fields ψ^I are doublets of $SU(2)$, $(-1)^H$ acts to change the sign of the fermion fields. That is

$$(-1)^H \psi^I (-1)^H = -\psi^I \tag{4.19}$$

or $\{(-1)^H, \psi^I\} = 0$. In the monopole sector, after expanding in zero modes we will then have $\{(-1)^H, b_0^a\} = 0$ and we must represent the action of $(-1)^H$ on the 2^{N_f} fold degenerate spectrum and impose the constraint (4.18). But this is a completely familiar problem. We can think of the b_0^a as gamma matrices and $(-1)^H$ as the analog of "γ^5 ". In other words, the spinor representation of $SO(2N_f)$ of dimension 2^{N_f} is reducible and splits into two irreducible representations, each of dimension 2^{N_f-1} with eigenvalues ± 1 under $(-1)^H$. Thus we learn from (4.18) that in the monopole sector there is a correlation between the electric charge of dyon states and their transformation properties under the global $SO(2N_f)$ symmetry. As discussed in [5] this is also required physically in order that one not make

states in monopole - antimonopole annihilation which do not occur in the perturbative spectrum.

One particularly interesting example of this phenomenon occurs for $N_f = 4$. Then by the above analysis, the fermion fields ψ^I carry electric charge $e/2$ and are in the eight-dimensional vector representation, 8_v, of the global $SO(8)$ symmetry (which we should really call $Spin(8)$ since there are spinors in the monopole sector.). On the other hand in the one monopole sector the spinor of dimension $2^4 = 16$ splits into two eight-dimensional spinor representations $16 \to 8_s + 8_c$ and from the constraint (4.18) we see that the neutral monopole (or evenly charged dyons) transforms as 8_s while the odd charged dyons transform as 8_c. Thus there seems to be a $Spin(8)$ triality as well as a possible electromagnetic duality in this theory, at least classically. In fact, when embedded into $N = 2$ supersymmetric gauge theory, this theory does appear to be self-dual with the $SL(2, Z)$ duality group extended to a triality action on $Spin(8)$ [5–7].

4.3 Monopoles coupled to isovector fermions

If we take the fermions in the adjoint representation then the index theorem (4.8) predicts two zero modes for a charge one monopole. Besides a doubling of the number of zero modes there is one other important difference from the isospinor case which involves the spin carried by the fermion zero modes. This can be understood as follows. From the discussion in sec. 2.1 we saw that the angular momentum generator for a symmetric monopole is

$$\mathbf{K} = \mathbf{L} + \mathbf{S} + \mathbf{T} \tag{4.20}$$

with $\mathbf{L} + \mathbf{S}$ the sum of orbital and spin terms generating the usual rotation group and $\mathbf{T}$ the $SU(2)$ generators. That is, the $SU(2)$ invariance group is a diagonal subgroup of the usual rotation group $SU(2)_R$ and the gauge group $SU(2)_G$.

Now isospinor fermions in the $\mathbf{2}$ of $SU(2)_G$ can have $K = 0$ since $\mathbf{2} \times \mathbf{2} = \mathbf{3} + \mathbf{1}$ and that is consistent with the fact that the zero modes (4.9) carry zero angular momentum as was implicitly assumed in the discussion in the previous section. But isovector fermions in the $\mathbf{3}$ necessarily have $K \neq 0$ since they transform in the product $\mathbf{3} \times \mathbf{2} = \mathbf{2} + \mathbf{4}$. Since the zero modes for adjoint fermions are two-fold degenerate but not four-fold degenerate the only possibility is that they carry spin 1/2. Thus we can write

$$\psi = a_{0,1/2}\psi_0^{1/2} + a_{0,-1/2}\psi_0^{-1/2} + \text{non-zero modes} \tag{4.21}$$

where the $\pm 1/2$ indicate the component of spin along say the z-axis. Following the previous analysis we then have a four-fold degenerate spectrum consisting of the states shown below. To simplify the notation I have dropped the zero subscript and written $\pm$ instead of $\pm 1/2$.

State	S_z
$\|\Omega\rangle$	0
$a_+^\dagger \|\Omega\rangle$	$\frac{1}{2}$
$a_-^\dagger \|\Omega\rangle$	$-\frac{1}{2}$
$a_+^\dagger a_-^\dagger \|\Omega\rangle$	0

(4.22)

So we see that by coupling adjoint fermions to monopoles we can give the monopoles spin. Remembering that one of the original problems with the Montonen-Olive proposal was the lack of monopole spin, this suggests that one way to cure the problem is to couple the monopoles to fermions in such a way as to obtain spin one monopoles. We will see in the next lecture that this is indeed possible.

We saw earlier that the bosonic collective coordinates of a single charge monopole, its location and dyon degree of freedom, could be thought of as arising from symmetries of the original Lagrangian which are broken by the monopole background (these symmetries are not broken by the vacuum, just by the monopole background). Since we have also found fermion zero modes or collective coordinates it is natural to wonder whether they can be viewed in the same way. In supersymmetric theories the answer is yes, the fermion zero modes (for charge one only) arise due to the supersymmetries which are unbroken in the vacuum but are broken by the monopole background. This is discussed in the following lecture.

4.4 Exercises for Lecture 4

E13. Find the unitary transformation which relates the gamma matrices used in this section to your favorite choice of gamma matrices.

E14. For N_f Dirac fermions in the doublet of $SU(2)$ we found N_f zero modes in a one monopole background which led to creation and annihilation operators a_0^i, $a_0^{i\dagger}$, $i = 1, 2, \cdots N_f$ obeying the anticommutation relations

$$\begin{aligned} \{a_0^i, a_0^j\} &= \{a_0^{i\dagger}, a_0^{j\dagger}\} = 0 \\ \{a_0^i, a_0^{j\dagger}\} &= \delta^{ij} \end{aligned} \tag{4.23}$$

a) Construct operators obeying the Lie algebra of $SU(N_f)$ in terms of the a_0^i and $a_0^{j\dagger}$.

b) Show that the monopole ground state has multiplicity 2^{N_f}. What representations of $SU(N_f)$ occur?

c) Show that one can in fact construct generators of $SO(2N_f)$ in terms of the a_0^i and $a_0^{j\dagger}$ and that the previous $SU(N_f)$ is embedded as $SO(2N_f) \supset SU(N_f)$ with $2N_f \to N_f + \bar{N_f}$ and that the monopole ground state transforms as the (reducible) spinor representation of $SO(2N_f)$ which decomposes as a sum of anti-symmetric tensor representations

$$2^{N_f} \to \sum_{M=0}^{N_f} \binom{N_f}{M}. \tag{4.24}$$

For assistance with this problem see [43].

E15. Construct the two isovector fermion zero modes $\psi_0^{\pm 1/2}$ for a charge one BPS monopole by solving the Dirac equation in this background. Construct the operator $\mathbf{K}$ and verify that the zero modes carry angular momentum $\pm 1/2$. For assistance see [38].

5 Lecture 5

5.1 Monopoles in $N = 2$ Supersymmetric Gauge Theory

In this lecture we will be considering theories with either $N = 2$ or $N = 4$ spacetime supersymmetries. Realistic (i.e. chiral) models of particle interactions have only $N = 1$ supersymmetry. There are theoretical reasons for discussing $N = 2$ and $N = 4$, the main one being that the dynamics of these theories is under much better control and this allows one to make statements about the spectrum which are valid non-perturbatively. One aspect of this which is discussed in the next section is the fact that the Bogomol'nyi bound follows as a consequence of the supersymmetry algebra for $N > 1$. Related to this is the fact that each supersymmetry relates field whose spin differs by 1/2. If we want all the fields we have discussed so far, A_μ, ψ, Φ with spins ranging from 1 to 0, to be related by supersymmetry than we require at least $N = 2$ supersymmetry. Monopoles in $N = 2$ supersymmetric gauge theories were first discussed in detail in reference [44].

If we count bosonic and fermionic degrees of freedom for the Lagrangian given by the sum of (3.13) and (4.1) (taking $\langle\Phi\rangle = 0$ for the moment) we have 3 physical bosonic degrees for each element of the adjoint representation of $SU(2)$ (two from the gauge fields and one from the Higgs field) and 4 fermionic degrees of freedom. Thus to have the possibility of a supersymmetric spectrum we must add an additional boson to the theory. This can be achieved by adding another Higgs field in the adjoint representation. This then gives the field content of $N = 2$ Super Yang-Mills theory. The Lagrangian, in component form, is given by

$$\begin{aligned}\mathcal{L}_{N=2} = \mathrm{Tr}\Big(&-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tfrac{1}{2}(D_\mu P)^2 - \tfrac{1}{2}(D_\mu S)^2 - \frac{e^2}{2}[S,P]^2 \\ &- \bar\psi\gamma^\mu D_\mu\psi - e\bar\psi[S,\psi] - e\bar\psi\gamma_5[P,\psi]\Big)\end{aligned} \tag{5.1}$$

where all fields are written as elements of the Lie algebra of $SU(2)$, i.e. $S = S^a T_a$ etc. and S and P are scalar Higgs fields. The Lagrangian (5.1) is invariant under the supersymmetry transformations

$$\begin{aligned}\delta A_\mu &= \bar\psi\gamma_\mu\alpha - \bar\alpha\gamma_\mu\psi \\ \delta P &= \bar\alpha\gamma_5\psi - \bar\psi\gamma_5\alpha \\ \delta S &= i\bar\alpha\psi - i\bar\psi\alpha \\ \delta\psi &= (\sigma^{\mu\nu}F_{\mu\nu} - i\not{D}S - \not{D}P\gamma_5 - i[P,S]\gamma_5)\alpha\end{aligned} \tag{5.2}$$

with α the Grassmann valued (Dirac) spinor supersymmetry parameter. Since the minimal $N = 1$ supersymmetry has one Majorana parameter and a Dirac spinor is equivalent to two Majorana spinors, (5.1) has $N = 2$ supersymmetry.

There is a potential term in the Lagrangian (5.1) but it has an exact flat direction whenever $[S, P] = 0$. Also, as in the simpler Lagrangians we considered with $V(\Phi) \equiv 0$, (5.2) is classically scale invariant and this scale invariance will be spontaneously broken by having a non-zero expectation value for the scalar fields. This is enough to ensure that at least classically there will be a massless Higgs field which is the dilaton of spontaneously broken scale invariance.

As before, we can impose as a boundary condition that $S^a S_a \to v^2$ as $r \to \infty$ thus breaking the gauge symmetry from $SU(2)$ down to $U(1)$. It should also be clear that we can trivially obtain a charge one BPS monopole solution in this theory using (2.1) with Φ^a replaced by S^a [9] so that we obtain a solution obeying the Bogomol'nyi equation

$$B_i = D_i S \tag{5.3}$$

Now following the earlier discussion we can ask whether this solution is invariant under the action of supersymmetry. Since we are starting with a classical solution with the fermion fields set to zero the supersymmetry variation of the bosonic fields is automatically zero. The supersymmetry variation of the fermion field ψ for this background is

$$\delta\psi = (\sigma^{\mu\nu} F_{\mu\nu} - i \not{D} S)\alpha \tag{5.4}$$

Now using (5.3) and writing

$$\Gamma_5 = i\gamma^0\gamma_5 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{5.5}$$

we obtain

$$\delta\psi = i\gamma^i B_i (1 - \Gamma_5)\alpha \tag{5.6}$$

Thus if we decompose α in terms of $\alpha_\pm = (1 \pm \Gamma_5)\alpha/2$ we see that the supersymmetries α_+ are unbroken in the monopole background while the α_- supersymmetries are broken. The variations (5.6) for the broken supersymmetries give zero energy Grassmann variations of the monopole solution, that is they are zero modes of the Dirac equation in the monopole background as can be seen by comparing (5.6) with the solution of Exercise E14.

5.2 The Bogomol'nyi Bound Revisited

Supersymmetry also gives important new insight into the Bogomol'nyi bound [45]. It is somewhat easier to work with the two independent Majorana components of the supersymmetry charge $Q_{\alpha i}$ with α being a spinor index and $i = 1, 2$ labeling the supersymmetry. The $N = 2$ supersymmetry algebra then takes the form

$$\{Q_{\alpha i}, \bar{Q}_{\beta j}\} = -i\delta_{ij}\gamma^\mu_{\alpha\beta} P_\mu + \delta_{\alpha\beta} U_{ij} + (\gamma_5)_{\alpha\beta} V_{ij} \tag{5.7}$$

where $U_{ij} = -U_{ji}$ and $V_{ij} = -V_{ji}$ are central terms which commute with the rest of the supersymmetry algebra. They can be evaluated in a specific theory by constructing the supercharges in terms of the underlying fields and then using the canonical (anti-)commutation relations. The calculations are detailed but straightforward and in the theory we are considering Witten and Olive found that

$$U_{ij} = \epsilon_{ij} v Q_e, \qquad V_{ij} = \epsilon_{ij} v Q_m \tag{5.8}$$

with (Q_e, Q_m) the electric and magnetic charge operators described previously.

[9] We can always choose $P^a = 0$ by a chiral rotation

It is then not hard to show that the supersymmetry algebra (5.7) implies the Bogomol'nyi bound $M \geq v\sqrt{Q_e^2 + Q_m^2}$. For example consider the case $Q_m = 0$. In the rest frame $P^\mu = (M, \mathbf{0})$ the supersymmetry algebra has the form

$$\{Q_{\alpha i}, Q_{\beta j}\} = \delta_{ij}\delta_{\alpha\beta}M + iv\epsilon_{ij}\gamma^0_{\alpha\beta}Q_e \tag{5.9}$$

The left hand side is positive definite while the second term on the right hand side has eigenvalues $\pm vQ_e$. We therefore conclude that $M \geq v|Q_e|$.

It is clear from the above argument that the bound is saturated precisely when one of the $Q_{\alpha i}$ is represented by zero, that is for states annihilated by at least one of the supersymmetry operators. This gives a beautiful relation between partially unbroken supersymmetry and BPS saturated states. In fact, we can turn the argument around and derive the Bogomol'nyi equation $B = DS$ by demanding that half of the supersymmetries (5.2) annihilate the monopole solution.

There is also a close connection between BPS saturated states and short representations of the $N = 2$ supersymmetry algebra. Roughly speaking what happens is the following. A massive representation of the $N = 2$ supersymmetry algebra is constructed by first going to the rest frame. The supersymmetry algebra then has the same form as a Clifford algebra and one can view linear combinations of the supercharges as creation and annihilation operators. The smallest representation of this algebra then has dimension 2^{2N} which is 16 for $N = 2$. On the other hand for massless representations one can go to a null frame and in this frame one finds that half of the supersymmetry charges anticommute to zero and are thus represented trivially. As a result representations have dimension $2^N = 4$. Now the $N = 2$ multiplet we started with consisting of (A_μ, ψ, S, P) has 8 states and consists of two irreducible massless representations of $N = 2$. When we take into account the Higgs mechanism some states eat others to get massive, but the total number of states does not change. We thus have 8 massive states. But this seems to contradict the previous analysis. The resolution of this is that in constructing massive representations the anticommutator of supercharges involves a particular combination of the mass and central charges V_{12}, U_{12}. For a special relation between the mass and central charges these combinations vanish and one again must represent only 1/2 as many supercharges non-trivially. This special relation if of course just the Bogomol'nyi bound. Further details on representations of $N = 2$ and the role of central charges can be found for example in [46].

5.3 Monopoles in $N = 4$ Supersymmetric Gauge Theory

The Super Yang-Mills theory with $N = 2$ supersymmetry we have been discussing so far describes a single vector multiplet with physical fields (A_μ, ψ, S, P). One can add to this theory hypermultiplets in an arbitrary representation of the gauge group. Hypermultiplets have a field content consisting of two Weyl fermions and four real scalars with quantum numbers so that they are in a real representation of the gauge group. If we consider a theory with one vector multiplet and one hypermultiplet, both in the adjoint representation of the gauge group and write down all possible renormalizable coupling consistent with $N = 2$ supersymmetry then it is known that the resulting theory in fact has $N = 4$ supersymmetry.

Another more fundamental way to think about the $N = 4$ theory uses the notion of dimensional reduction from a higher dimensional theory [47]. $N = 1$ supersymmetric Yang-Mills theory with field content consisting only of gauge fields and their supersymmetric gaugino partners is possible only in $D = 3, 4, 6$ and 10 spacetime dimensions. In order to have the correct matching of physical degrees of freedom on must impose conditions of the fermion fields and supersymmetries. In $D = 3$ dimensions the supersymmetry and fermion fields must be Majorana, in $D = 4$ Majorana or Weyl, which are equivalent, in $D = 6$ Weyl and there is no Majorana condition, and finally in $D = 10$ on must impose the Majorana and Weyl conditions simultaneously.

Thus in ten dimensions we start with a spinor λ in the adjoint representation of some group G (which we choose to be $SU(2)$ for simplicity) and obeying both a chirality condition and Majorana condition:

$$(1 + \Gamma_{11})\lambda = 0, \qquad \bar{\lambda} = \lambda^T C \tag{5.10}$$

with C the charge conjugation matrix. The $N = 1$ Lagrangian in ten dimensions with $A, B = 0, 1, 2 \ldots 9$ is then

$$\mathcal{L} = \mathrm{Tr}\left(-\frac{1}{4}F_{AB}F^{AB} - \frac{1}{2}\bar{\lambda}\gamma^A D_A \lambda\right) \tag{5.11}$$

The dimensional reduction of this Lagrangian is carried out in detail in [38] specifically for the purpose of analyzing the monopole spectrum and the details will not be repeated here. However some general features should be pointed out. From a group theoretical point of view the dimensional reduction reduces the Lorentz group via $SO(9,1) \supset SO(3,1) \times SO(6)$. In the dimensional reduction we discard all dependence on six of the coordinates, as a result the $SO(6)$ part of the ten-dimensional Lorentz group will acts as a global symmetry of the $N = 4$ theory. The gauge fields and fermion fields transform under this reduction as $10 \to (4,1) + (1,6)$ and $16 \to (2_+, 4_+) + (2_-, 4_-)$ respectively where the subscript indicate the chirality. As a result the four-dimensional spectrum will consist of a gauge field, six scalars Φ_a in the 6 of $SO(6)$ and four Weyl spinors λ_i transforming as a 4 of $SO(6)$ (or to be precise $Spin(6) \equiv SU(4)$). The resulting Lagrangian can be written in a variety of forms, not all of which make the $SO(6)$ symmetry manifest. Probably the most elegant formalism uses the fact that the 6 of $SU(4)$ arises in the antisymmetric product $(4 \times 4)_A$ to write the six scalar fields in terms of an antisymmetric complex matrix Φ_{ij}, $i, j = 1 \ldots 4$ obeying the condition $(\Phi_{ij})^\dagger = \Phi^{ij} = (1/2)\epsilon^{ijkl}\Phi_{kl}$. The Lagrangian is then in two component form

$$\begin{aligned}\mathcal{L}_{N=4} = \mathrm{Tr}\Big(&-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\lambda_i\sigma^\mu D_\mu\bar{\lambda}^i - \tfrac{1}{2}D_\mu\Phi_{ij}D^\mu\Phi^{ij} \\ &+ i\lambda_i[\lambda_j, \Phi^{ij}] + i\bar{\lambda}^i[\bar{\lambda}^j, \Phi_{ij}] + \frac{1}{4}[\Phi_{ij}, \Phi_{kl}][\Phi^{ij}, \Phi^{kl}]\Big)\end{aligned} \tag{5.12}$$

As in our discussion of the $N = 2$ theory, it is clear that the $N = 4$ theory is classically scale invariant, that the potential has flat directions, and that we can give expectation values to the scalars which will break the $SU(2)$ gauge symmetry to $U(1)$ and spontaneously break scale invariance. For example a simple choice would be to take only the field Φ_{12} to have a non-zero expectation value. We could then embed the BPS solution into this theory by replacing the Higgs field Φ in the BPS solution by Φ_{12}.

Giving a non-zero expectation value to Φ_{12} not only breaks the $SU(2)$ gauge symmetry to $U(1)$, it also breaks the classical scale invariance and also spontaneously breaks the global $SO(6)$ symmetry to $SO(5)$. It is known that $N = 4$ Super Yang-Mills theory is a finite theory with vanishing beta function and thus an exact quantum scale invariance. The scalar spectrum after gauge symmetry breaking will therefor consist of one massless scalar (the dilaton) which is the Nambu-Goldstone (NG) boson of spontaneously broken scale invariance and five massless scalars which are the NG bosons of the spontaneously broken $SO(6)$ global symmetry.

From (5.12) we see that the fermions have the standard Yukawa and gauge couplings to the fields appearing in the BPS monopole solution. It should thus be clear that the charge one BPS monopole embedded in the $N = 4$ theory has twice as many fermion zero modes as we had for the pure $N = 2$ theory since there are now the equivalent of two Dirac fermions (i.e. four Weyl fermions) in the adjoint representation of $SU(2)$. Following the discussion around (4.21) we will now have fermion zero modes in the one monopole sector $a^n_{0,\pm 1/2}$ with $n = 1, 2$ labeling the two fermion fields. Dropping the zero subscript and writing $\pm$ for $\pm 1/2$ as before the spectrum will then consist of the states

$$\begin{array}{rc} \text{State} & S_z \\ |\Omega\rangle & 0 \\ a^{n\dagger}_{\pm}|\Omega\rangle & \pm\frac{1}{2} \\ a^{n\dagger}_{-}a^{m\dagger}_{+}|\Omega\rangle & 0 \\ a^{1\dagger}_{+}a^{2\dagger}_{+}|\Omega\rangle & 1 \\ a^{1\dagger}_{-}a^{2\dagger}_{-}|\Omega\rangle & -1 \\ a^{1\dagger}_{\mp}a^{2\dagger}_{\mp}a^{n\dagger}_{\pm}|\Omega\rangle & \mp\frac{1}{2} \\ a^{1\dagger}_{+}a^{2\dagger}_{+}a^{1\dagger}_{-}a^{2\dagger}_{-}|\Omega\rangle & 0 \end{array} \tag{5.13}$$

for a total of 16 states, 8 bosons, 6 with spin 0 and 2 with spin ± 1 and 8 fermions with spin $\pm 1/2$. This is the same as the content of the gauge super multiplet of $N = 4$ Yang-Mills theory.

Thus in $N = 4$ gauge theory we finally see that it is possible to obtain monopoles of spin one and in fact the monopole supermultiplet and the gauge supermultiplet are the same in this theory. This in fact is not so surprising, there is a unique multiplet in $N = 4$ gauge theory which does not contain spin greater than one.

5.4 Supersymmetric Quantum Mechanics on $\mathcal{M}_k$

We argued earlier that we can think of the fermion zero modes as Grassmann collective coordinate for the monopole moduli space. In the absence of fermion fields the collective coordinate expansion with background fields $A^i(x^j, z^\alpha(t))$, $\Phi(x^j, z^\alpha(t))$ leads to a low-energy effective action by substituting into the four-dimensional action and integrating over R^3 to obtain

$$S_{\text{eff}} = \int dt\, \mathcal{G}_{\alpha\beta}\dot{z}^\alpha \dot{z}^\beta \tag{5.14}$$

with $\mathcal{G}$ the metric on the monopole moduli space. In other words, at very low energies the field theory about the monopole background can excite only a finite number of degrees of freedom, those that correspond to motion along the moduli space, and thus we can reduce the dynamics to quantum mechanics.

We would like to add fermion zero modes to this picture. We can do this by also expanding the fermion fields as

$$\psi = \sum_\alpha \lambda^\alpha(t)\psi_{0\alpha}(x,z) \tag{5.15}$$

where the $\psi_{0\alpha}$ are the fermion zero modes, treated as real numbers, and the $\lambda^\alpha(t)$ are Grassmann valued fermion collective coordinates. One can include these in the quantum mechanical effective action by carrying out the same procedure as before. The details are somewhat subtle however and will not be presented here. For details see [48] for the analysis in $N = 2$ theories and [49] in $N = 4$ theories. However the answer is not surprising. The quantum mechanical action is extended to the action for a supersymmetric quantum mechanics. In the $N = 2$ case there are four real supersymmetries in spacetime which are unbroken in the monopole background. As a result we expect an action with $N = 4$ world-line supersymmetry. This action is

$$S_{\text{eff}} = \tfrac{1}{2}\int dt\, \mathcal{G}_{\alpha\beta}(\dot{z}^\alpha \dot{z}^\beta + 4i\lambda^{\dagger\alpha} D_t \lambda^\beta) + \text{const.} \tag{5.16}$$

where

$$D_t\lambda^\alpha = \frac{d\lambda^\alpha}{dt} + \Gamma^\alpha{}_{\beta\gamma}\frac{dz^\beta}{dt}\lambda^\gamma \tag{5.17}$$

is the covariant derivative acting on the spinor λ. The nomenclature for supersymmetry in quantum mechanics is a bit confusing. Originally actions with two component spinors and one supersymmetry were constructed and the supersymmetry was referred to as $N = 1$. It was later realized that one could also have one supersymmetry with one component spinors and this was unfortunately called $N = 1/2$ supersymmetry. With this nomenclature the action (5.16) might be said to possess $N = 4 \times 1/2$ supersymmetry. The presence of four supersymmetries requires that the moduli space be a hyperkahler manifold. This means $\mathcal{M}_k$ has three complex structures $\mathcal{J}^m$ which obey

$$\mathcal{J}^m{}_\beta{}^\alpha \mathcal{J}^n{}_\alpha{}^\gamma = -\delta^\gamma_\beta \delta^{mn} + \epsilon^{mn}{}_p \mathcal{J}^p{}_\beta{}^\gamma \tag{5.18}$$

and the action has $N = 4 \times 1/2$ supersymmetry with supersymmetry transformations

$$\begin{aligned} \delta z^\alpha &= i\beta_4 \lambda^\alpha + i\beta_m \lambda^\beta \mathcal{J}^m{}_\beta{}^\alpha \\ \delta\lambda^\alpha &= -\dot{z}^\alpha \beta_4 - \beta_m \dot{z}^\beta \mathcal{J}^m{}_\beta{}^\alpha. \end{aligned} \tag{5.19}$$

The BPS monopole moduli space can be shown to be hyperkahler, independent of supersymmetry [16], but to a physicist, supersymmetry provides the simplest explanation of this fact.

If we choose one of the complex structures to introduce complex coordinates on $\mathcal{M}_k$ then we have the canonical anti-commutation relations

$$\{\lambda^\alpha, \lambda^{\bar{\beta}}\} = \delta^{\alpha\bar{\beta}}. \tag{5.20}$$

The $\lambda^{\bar{\alpha}}$ therefore act as creation operators and we thus have a spectrum of states of the form

$$f_{\bar{\alpha}_1\cdots\bar{\alpha}_p}\lambda^{\bar{\alpha}_1}\cdots\lambda^{\bar{\alpha}_p}|\Omega\rangle. \tag{5.21}$$

These state are in one to one correspondence with holomorphic $(0,p)$ forms on $\mathcal{M}_k$

$$|f\rangle = f_{\bar{\alpha}_1\cdots\bar{\alpha}_p}\lambda^{\bar{\alpha}_1}\cdots\lambda^{\bar{\alpha}_p}|\Omega\rangle \Longleftrightarrow f_{\bar{\alpha}_1\cdots\bar{\alpha}_p}dz^{\bar{\alpha}_1}\wedge\cdots dz^{\bar{\alpha}_p}. \tag{5.22}$$

In the reduction of $N=4$ Super Yang-Mills theory one obtains a Lagrangian with $N=4\times 1$ supersymmetry (twice as much supersymmetry as the $N=2$ theory) given by

$$S_{\text{eff}} = \tfrac{1}{2}\int dt\, \mathcal{G}_{\alpha\beta}(\dot{z}^\alpha\dot{z}^\beta - \bar{\psi}^\alpha\gamma^0 D_t\psi^\beta) + \frac{1}{6}R_{\alpha\beta\gamma\delta}(\bar{\psi}^\alpha\psi^\gamma)(\bar{\psi}^\beta\psi^\delta) \tag{5.23}$$

where now ψ^α is a two-component spinor rather than a one-component object as in (5.16). As a result of this doubling one now finds that the Hilbert space of states is the same as the space of all differential forms on $\mathcal{M}_k$ and that the Hamiltonian is the Laplacian acting on forms. For further details of this correspondence see [50, 51].

5.5 Exercises for Lecture 5

E16. Verify that $\delta\chi = i\gamma^i B_i\alpha_-$ is a zero-mode of the Dirac equation and that this agrees with what you found in Exercise 12.

E17. Construct the supercharges Q, Q^* for the theory described by (5.16). Show that with the above correspondence between states and forms the supercharges and Hamiltonian are given by

$$\begin{aligned} Q^* &\Leftrightarrow \bar{\partial} \\ Q &\Leftrightarrow \bar{\partial}^\dagger \\ H = \{Q, Q^*\} &\Leftrightarrow \bar{\partial}^\dagger\bar{\partial} + \bar{\partial}\bar{\partial}^\dagger = \tfrac{1}{2}(dd^\dagger + d^\dagger d) \end{aligned} \tag{5.24}$$

where the latter equality uses the fact the $\mathcal{M}_k$ is Kahler. Thus the Hamiltonian is just the Laplacian acting on forms.

E18. Show that the $N=2$ Yang-Mills action (5.1) can be derived by dimensional reduction of $N=1$ Yang-Mills theory in six spacetime dimensions.

6 Lecture 6

6.1 Implications of S-duality

It is time to take stock of where we are in the search for theories which may exhibit an exact electromagnetic duality. We have seen that this cannot be the case in pure $SO(3)$ gauge theory or in $N=2$ Yang-Mills theory because the monopole does not have spin one and thus cannot be dual to the W boson [10]. On the other hand in $N=4$ Yang-Mills theory

[10] This does not rule out a duality relating monopoles to fermion matter fields in $N=2$ theories, strong evidence for such a duality was found in [5] but a full discussion of this would lead us to far afield.

there is only one supermultiplet which contains only spin ≤ 1 and we have seen that the monopoles and gauge bosons both lie in this supermultiplet.

In addition, although we will not discuss it in these lectures, quantum corrections in $N = 4$ are under very precise control. In fact it is known that this theory has vanishing beta function, both perturbatively and non-perturbatively. This ensures that the flat direction in the potential we are utilizing remains in the full theory and that BPS states constructed at weak coupling continue to exist and evolve smoothly to states at strong coupling.

We have thus addressed the first two objections to the Montonen-Olive proposal. We are still faced with finding a non-trivial way to check the proposed duality without having to compute directly at strong coupling. This is where the extension of duality to $SL(2, Z)$ plays a central role as was first appreciated by Sen. We will make one assumption, namely that the state with $(n_e, n_m) = (1, 0)$ (the W^+ boson) exists at all values of the coupling τ with a degeneracy of 16 corresponding to the 16 states in the short vector representation of $N = 4$ supersymmetry. This is an extremely mild assumption. We have already argued that the dimension of the representation cannot change as parameters of the theory are varied and we know that such a state exists at weak coupling as a BPS saturated state. The one known mechanism by which BPS saturated states can disappear requires that the lattice spanned by the electric and magnetic charges degenerate at some value of τ and in this theory that is ruled out by the non-renormalization theorems.

Given the existence of the $(1, 0)$ state $SL(2, Z)$ duality requires the existence of all the $SL(2, Z)$ images of this state. Since a $SL(2, Z)$ transformation acts on this state as

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} \tag{6.1}$$

duality requires the existence of states with $(n_e, n_m) = (a, c)$ with the same degeneracy of 16. Furthermore, since $ad - bc = 1$ it follows that a and c are relatively prime, $(a, c) = 1$. Also, by starting at a value of τ corresponding to strong coupling and then performing the duality transformation (6.1) these states must exist at weak coupling and thus should appear in a semi-classical analysis of the spectrum.

For $c = 1$ we require states $(a, 1)$ for arbitrary integer a. These are just the dyonic excitations of the single charge BPS monopole and from the previous analysis we see that such states do exist with the correct multiplicity. Furthermore these states are BPS states as is demonstrated in [48]. For $c = 2$ we require states $(a, 2)$ with a odd, again with a degeneracy of 16. Previous to Sen's analysis such states were not known to exist.

Constructing these states in the full $N = 4$ supersymmetric field theory would be very difficult. Luckily the question can be reduced to construction of bound states in the moduli space approximation. To see why this is the case consider a bound state with electric charge one. A BPS monopole state with $(n_e, , n_m) = (0, 1)$ has mass $M_{(0,1)} = vg$ while a dyon state with charge $(1, 1)$ has mass $M_{(1,1)} = v\sqrt{e^2 + g^2}$. A BPS bound state of charge $(2, 1)$ on the other hand has mass $M_{(2,1)} = v\sqrt{4g^2 + e^2}$. At weak coupling the binding energy is thus

$$M_{(2,1)} - M_{(1,1)} - M_{(1,0)} \sim ve(e/4g) \ll ve. \tag{6.2}$$

Since this is much less than the W mass we should be able to study the existence of this bound state in the moduli space approximation. The same argument applies to states of greater electric charge at sufficiently weak coupling.

In the following we will see how the existence of these states follows from a careful analysis of supersymmetric quantum mechanics on $\mathcal{M}_2$. Evidence for the existence of states with arbitrary (a, c)=1 can be found in [52, 53]. What about states with charges $(0, n_m)$ or $(n_e, 0)$? At weak coupling we know that there are no electric charge two bound states in the spectrum, there are BPS states in the theory with charge $(n_e, 0)$ but these are not distinct from the multi particle continuum of states. Similarly, although there are charge $(0, n_m)$ monopoles, our analysis will show (for $n_m = 2$) that these are not normalizable bound states of n_m single charge monopoles but just part of the continuum of states of n_m single charge monopoles [11].

6.2 The Two-monopole Moduli Space From Afar

As we will see in the following section, the metric on the two-monopole moduli space can be constructed exactly and the spectrum of supersymmetric quantum mechanics on this space can thus be determined by explicit calculations. However instead of proceeding directly to this analysis I would like to discuss briefly a description of the asymptotic form of the two monopole moduli space due to Manton [55]. There are two reasons for doing this. First, it brings out the physics of the moduli space in a direct way that is not obscured by difficult mathematics or special functions. Second, this approximate description has played a role in studies of the multi-monopole moduli space [56] and other problems involving moduli spaces.

We begin with the fact that the BPS monopole has magnetic charge and dilaton charge both equal to g with the definition of dilaton charge given in (2.11). At large distances from the monopole we can summarize this fact by writing down a point like interaction between the monopole and the photon and dilaton fields with equal strength interactions. We begin with the interaction with the photon field. Instead of working with the conventional vector potential A_μ as in the previous sections it is useful to introduce a dual potential $\tilde{A}^\mu = (\tilde{A}^0, \tilde{\mathbf{A}})$ defined by $\tilde{F} = d\tilde{A}$ in order to describe the field of a point monopole. We can then couple a point monopole of mass M to the photon by mimicking the interaction of a electrically charged point particle [12]:

$$S_{\tilde{A}} = \int dt \left(-M\sqrt{1-\mathbf{v}^2} - g\tilde{A}^0 + g\mathbf{v}\cdot\tilde{\mathbf{A}} \right). \tag{6.3}$$

When this is coupled to the electromagnetic action

$$S_{EM} = -\frac{1}{4}\int d^4x F_{\mu\nu}F^{\mu\nu} = -\frac{1}{4}\int d^4x \tilde{F}_{\mu\nu}\tilde{F}^{\mu\nu} \tag{6.4}$$

[11] This is in distinction to the situation for fundamental strings or some string, D-brane configurations where there are discrete states with multiple charge which should thought of as bound states and distinguished from the multiparticle continuum [54].

[12] The form of the action below is not manifestly covariant since it involves the time t as a parameter rather than the proper time τ. It does of course lead to the correct covariant equations of motion [57] and is more convenient for our purposes.

one finds as required that a monopole at rest at the origin gives rise to a Coulomb magnetic field, and the action of this field on a second monopole gives rise to the standard Coulomb repulsion between like sign monopoles. Now since the theory also includes a massless dilaton field with action

$$S_{dil} = -\frac{v^2}{2}\int d^4x\, \partial_\mu \mathcal{D}\partial^\mu \mathcal{D} \tag{6.5}$$

we must also include the coupling to the dilaton to obtain the correct force law. The coupling to the dilaton is dictated by the fact that a shift of the dilaton is equivalent to a shift in the mass of the monopole. This is the statement of spontaneously broken scale invariance. We can thus generalize (6.3) to

$$S_{\tilde{A},\mathcal{D}} = \int dt \left((-M + v\mathcal{D})\sqrt{1-\mathbf{v}^2} - g\tilde{A}^0 + g\mathbf{v}\cdot\tilde{\mathbf{A}} \right) \tag{6.6}$$

and now if we compute the net force between two stationary monopoles we find that the Coulomb repulsion is precisely cancelled by the dilaton attraction.

Now we can ask what happens if the monopoles move relative to each other at low velocities. At small velocities and at large impact parameter the interactions will still be mediated by exchange of massless particles so the previous description should suffice. If the first monopole is moving at velocity $\mathbf{v}_1 \ll 1$ a standard computation of the Lienard-Wiechert potentials and dilaton field to first order in the velocity gives

$$\begin{aligned} \tilde{A}^0 &= \frac{g}{4\pi r}, \\ \tilde{\mathbf{A}} &= \frac{g}{4\pi r}\mathbf{v}_1, \\ v\mathcal{D} &= \frac{g}{4\pi r}\sqrt{1-\mathbf{v}_1^2}, \end{aligned} \tag{6.7}$$

with $\mathbf{r}$ the relative separation between the monopoles. If we now substitute these fields into the Lagrangian for the second monopole and separate out the center of mass motion we are left with an action which governs the relative motion of the two monopoles:

$$S_{rel} = \int dt \left(\frac{M}{4} - \frac{g^2}{8\pi r} \right) \frac{d\mathbf{r}}{dt}\cdot\frac{d\mathbf{r}}{dt}. \tag{6.8}$$

The action (6.8) has no potential term, and a spatially varying kinetic term. We can interpret it as saying that the relative motion is geodesic motion for a metric on R^3 given by $ds^2 = U(r)d\mathbf{r}\cdot d\mathbf{r}$ with $U(r) = 1 - g^2/(2\pi M r)$. Note that so far everything we have done could also have been done for a electrically charged W boson in this theory, the low velocity motion of purely electrically or magnetically charged particles in this theory is equivalent to geodesic motion. Physically, the forces due to photon and dilaton exchange no longer cancel at non-zero velocity due to the different retardation effects for spin zero and spin one exchange.

Now for monopoles there is a natural generalization of this result. We saw previously that the classical single monopole moduli space has the form $R^3 \times S^1$ where the "velocity" on the S^1 factor determines the dyons electrical charge. Thus to generalize the above result

to scattering of dyons we should include the electrical charge of the dyons and view this as a velocity in some additional coordinate on S^1. The analysis is slightly complicated by the fact that one must use both A and $\tilde{A}$ in the computation, but is essentially a straightforward generalization of what we have done. In carrying out the computation one should remember that the dilaton must couple to $\sqrt{e^2+g^2}$ since this is the mass of the dyon from the BPS bound. The analysis is carried out in [55] with the result that the relative motion of two dyons with electric and magnetic charges (e_1, g_1), (e_2, g_2) and relative electric charge $e = e_2 - e_1$ is given by

$$S_{rel} = \int dt \left(\frac{M}{4} - \frac{g^2}{8\pi r} \right) \frac{d\mathbf{r}}{dt} \cdot \frac{d\mathbf{r}}{dt} + \frac{eg}{4\pi} \frac{d\mathbf{r}}{dt} \cdot \boldsymbol{\omega} + \frac{e^2}{8\pi r} \tag{6.9}$$

where $\boldsymbol{\omega}$ is the Dirac monopole potential determined by $\nabla \times \boldsymbol{\omega} = \mathbf{r}/r^3$. From the previous discussion we would like to regard the electric charge e as the velocity along a S^1 governing the relative charge of the two monopoles. If we call this fourth relative coordinate χ with $e \sim \dot{\chi}$ then the equations of motion following from (6.9) are equivalent to the equations for geodesic motion in the metric

$$ds^2 = U(r) d\mathbf{r} \cdot d\mathbf{r} + \frac{g^2}{2\pi M U(r)} (d\chi + \boldsymbol{\omega} \cdot d\mathbf{r})^2. \tag{6.10}$$

The metric (6.10) is known in the relativity literature as the Taub-NUT metric with negative mass. From our derivation, we only expect it to agree with the exact metric on the two monopole moduli space for monopole separations large compared to the inverse W mass where the Dirac monopole approximation is valid.

6.3 The Exact Two-monopole Moduli Space

Although the previous analysis gives a nice physical picture of the metric on the two monopole moduli space at large separation, we need the full metric in order to provide a precise test of duality. The exact two-monopole moduli space has been determined by Atiyah and Hitchin using the fact that it has $SO(3)$ isometry arising from rotational invariance, the fact that in four dimensions hyperkahler implies self-dual curvature, and the fact that the metric is known to be complete. This reduces the problem to an analysis of specific differential equations which can then be solved in terms of elliptic functions. Luckily we will not need to investigate the detailed form of the metric. A good reference for what follows is [19].

The two monopole moduli space has the form

$$\mathcal{M}_2 = R^3 \times \left(\frac{S^1 \times \mathcal{M}_2^0}{Z_2} \right) \tag{6.11}$$

where the R^3 factor is the overall center of mass of the system and the S^1 factor describes the overall dyon rotator degree of freedom. The reduced moduli space $\mathcal{M}_2^0$ is four-dimensional and when the two monopoles are far apart one can think of the coordinate on $\mathcal{M}_2^0$ as being the relative separation of the monopoles and the relative orientation

of the dyon degrees of freedom. Thus in this asymptotic region we have

$$\begin{aligned} \chi &= \tfrac{1}{2}(\chi_1 + \chi_2) \\ \mathbf{X} &= \tfrac{1}{2}(\mathbf{x}_1 + \mathbf{x}_2) \end{aligned} \tag{6.12}$$

as coordinates on $S^1 \times R^3$ and

$$\begin{aligned} \psi &= \tfrac{1}{2}(\chi_1 - \chi_2) \\ \mathbf{x} &= \tfrac{1}{2}(\mathbf{x}_1 - \mathbf{x}_2) \end{aligned} \tag{6.13}$$

as coordinates on $\mathcal{M}_2^0$ with $(\mathbf{x}_1, \chi_1)$ the collective coordinates of monopole one and similarly for monopole two. The Z_2 identification in (6.11) arises because a 2π rotation of one of the dyon degrees of freedom leads to the same monopole configuration. Explicitly it is given by the transformation

$$I_1 : \quad \psi \to \psi + \pi, \quad \chi \to \chi + \pi. \tag{6.14}$$

The explicit metric on $\mathcal{M}_2^0$ is given by

$$ds^2 = f(r)^2 dr^2 + a(r)^2 (\sigma_1^R)^2 + b(r)^2 (\sigma_2^R)^2 + c(r)^2 (\sigma_3^R)^2 \tag{6.15}$$

where the σ_i are left-invariant one-forms on $SO(3) = S^3/Z_2$[13]. In one particular basis they are given by

$$\begin{aligned} \sigma_1^R &= -\sin\psi d\theta + \cos\psi \sin\theta d\phi \\ \sigma_2^R &= \cos\psi d\theta + \sin\psi \sin\theta d\phi \\ \sigma_3^R &= d\psi + \cos\theta d\phi \end{aligned} \tag{6.16}$$

with with $0 \le \theta \le \pi$, $0 \le \phi \le 2\pi$, $0 \le \psi < 2\pi$. The angles are further restricted under the identification of the discrete right isometry [19]

$$(\phi, \theta, \psi) I_x = (\pi + \phi, \pi - \theta, -\psi). \tag{6.17}$$

Note that we can equivalently let the range of ψ be $0 \le \psi < 4\pi$ and then divide out by I_x.

We will follow [19] in choosing $f(r) = -b(r)/r$. The radial functions $a(r)$, $b(r)$ and $c(r)$ are given explicitly in [16]. Here we only need the asymptotic forms. Near $r = \pi$ they take the form

$$\begin{aligned} a(r) &= 2(r-\pi)\left\{1 - \frac{1}{4\pi}(r-\pi)\right\} + \dots \\ b(r) &= \pi\left\{1 + \frac{1}{2\pi}(r-\pi)\right\} + \dots \\ c(r) &= -\pi\left\{1 - \frac{1}{2\pi}(r-\pi)\right\} + \dots . \end{aligned} \tag{6.18}$$

Introducing appropriate Euler angles, it can be shown that after the identification by I_x the metric is smooth near $r = \pi$ and that $r = \pi$ is an S^2 or bolt [19]. Near infinity, $r \to \infty$,

[13] The metric (6.15) thus has $SO(3)$ isometry since it is invariant under the (left) action of $SO(3)$. It may seem odd at first sight that the metric has $SO(3)$ isometry is spite of having different radial functions multiplying each of the terms in (6.15). However if all radial functions were equal the metric would have $SO(4) = SU(2) \times SU(2) \sim SO(3) \times SO(3)$ isometry given by both the left and right actions of $SO(3)$.

the functions take the form

$$\begin{aligned} a(r) &= r\left(1-\frac{2}{r}\right)^{1/2}+\dots \\ b(r) &= r\left(1-\frac{2}{r}\right)^{1/2}+\dots \\ c(r) &= -2\left(1-\frac{2}{r}\right)^{-1/2}+\dots, \end{aligned} \tag{6.19}$$

where the neglected terms fall off exponentially with r. It can be shown that this asymptotic metric is equivalent to the Taub-NUT metric (6.10).

6.4 Duality and Sen's Two-form

We now want to use the metric on the two monopole moduli space to partially test the predictions of S duality following Sen's original analysis. In particular, S duality predicts the existence of BPS saturated bound states with magnetic charge 2 and odd electric charge.

In the moduli space approximation BPS states are supersymmetric ground states, and these in term correspond to harmonic forms on the moduli space as discussed at the end of the previous lecture. The fact that we are looking for a bound state means that the wave function or form on the relative moduli space must be normalizable, that is in L^2.

In quantizing the theory on $\mathcal{M}_2$ we will obtain 16 fold degenerate states from the wedge product of the 16 harmonic (constant) forms on $R^3 \times S^1$ with L^2 harmonic forms on $\mathcal{M}_2^0$. S-duality predicts that we have precisely this degeneracy so it requires the existence of a unique L^2 harmonic form on $\mathcal{M}_2^0$. Furthermore, for the corresponding states to have odd electric charge this form must be odd under the Z_2 action (6.14) . Now since the Hodge dual of a harmonic form is also harmonic, it follows that we can get a unique harmonic form only if the form is self-dual or anti-self-dual.

With this information it is then straightforward to write down the candidate form. It is given by the ansatz

$$\omega = F(r)\left(d\sigma_1 - \frac{fa}{bc}dr\wedge\sigma_1\right). \tag{6.20}$$

Note that this is anti-self-dual by construction. Demanding that ω be harmonic yields the equation

$$\frac{dF}{dr} = -\frac{fa}{bc}F. \tag{6.21}$$

An analysis of this equation at infinity and at the "bolt" shows that the form ω is normalizable and well behaved at the bolt [2]. Furthermore this is the unique such form. This thus establishes the existence of precisely the BPS bound states which are required by S duality in the two monopole sector.

This analysis also shows that there are no such BPS bound states in $N = 2$ super Yang-Mills theory without matter [51]. In the $N = 2$ theory supersymmetric ground states are holomorphic forms on the moduli space, but the form (6.20) being anti-self-dual is a

$(1,1)$ form and thus not holomorphic. Of course in the $N = 2$ theory there was no reason to expect such bound states since the theory is not S dual.

We have shown that such states exist mathematically, but one might wonder what the physics is behind this result and whether there is some simple explanation why we found bound states for $N = 4$ but not for $N = 2$. I believe the answer has to do with spin dependent forces. Once the monopoles have spin there will be additional long range forces (e.g. spin-orbit and spin-spin) besides those considered in Manton's analysis of the asymptotic moduli space. Depending on the magnitude of the spin these spin dependent forces can lead to bound states which would not otherwise exist. So for example, without supersymmetry, bound states of the basic Prasad-Sommerfield theory with vanishing potential would correspond to L^2 harmonic functions rather than forms on the two monopole moduli space and we have seen that these do not exist. With $N = 2$ supersymmetry the absence of a four fermion term in the supersymmetric quantum mechanics indicates a cancellation of the spin-spin forces between vector and scalar exchange. It is only when we get to $N = 4$ supersymmetry and spin one monopoles that the spin dependent forces can lead to new BPS saturated bound states.

6.5 Exercises for Lecture 6

E19. Analyze the asymptotic form of the two monopole moduli space given the asymptotic formulae for $a(r)$, $b(r)$, $c(r)$ and $f(r)$. Show that the asymptotic metric can be put in the from of the Taub-NUT metric (6.10).

E20. The following problem is an extended exercise in the geometry of the $SU(2)$ group manifold, a.k.a. S^3. Parameterize the three sphere by Euler angles and write a general $SU(2)$ rotation as

$$\begin{aligned} U(\phi,\theta,\psi) =& U_z(\phi)U_y(\theta)U_z(\psi) \\ =& \begin{pmatrix} \cos\frac{\theta}{2}e^{i\frac{(\psi+\phi)}{2}} & \sin\frac{\theta}{2}e^{-i\frac{(\psi-\phi)}{2}} \\ -\sin\frac{\theta}{2}e^{i\frac{(\psi-\phi)}{2}} & \cos\frac{\theta}{2}e^{-i\frac{(\psi+\phi)}{2}} \end{pmatrix} \end{aligned} \tag{6.22}$$

a) By expanding the one-form $U^{-1}dU$ in the basis $i\tau_i/2$ with τ_i the Pauli matrices construct the left-invariant or "right" one-forms σ_i^R. Similarly, expand dUU^{-1} to obtain a set of right-invariant "left" one-forms σ_i^L. The one-forms $\sigma_i^{R,L}$ are dual to left (right) invariant vector fields ξ_i^R (ξ_i^L) which generate right (left) group actions.

b) Construct explicitly the dual vector fields satisfying $\langle \xi_i^R, \sigma_j^R \rangle = \delta_{ij}$ and $\langle \xi_i^L, \sigma_j^L \rangle = \delta_{ij}$ and show that they are given by

$$\begin{aligned} \xi_1^R &= -\cot\theta\cos\psi\partial_\psi - \sin\psi\partial_\theta + \frac{\cos\psi}{\sin\theta}\partial_\phi \\ \xi_2^R &= -\cot\theta\sin\psi\partial_\psi + \cos\psi\partial_\theta + \frac{\sin\psi}{\sin\theta}\partial_\phi \\ \xi_3^R &= \partial_\psi \end{aligned} \tag{6.23}$$

and

$$\begin{aligned}\xi_1^L &= -\frac{\cos\phi}{\sin\theta}\partial_\psi + \sin\phi\partial_\theta + \cot\theta\cos\phi\partial_\phi \\ \xi_2^L &= \frac{\sin\phi}{\sin\theta}\partial_\psi + \cos\phi\partial_\theta - \cot\theta\sin\phi\partial_\phi \\ \xi_3^L &= \partial_\phi.\end{aligned} \tag{6.24}$$

c) Show that the left and right invariant one forms satisfy the Maurer-Cartan equations

$$\begin{aligned}d\sigma_i^R &= \frac{1}{2}\epsilon_i{}^{jk}\sigma_j^R \wedge \sigma_k^R \\ d\sigma_i^L &= -\frac{1}{2}\epsilon_i{}^{jk}\sigma_j^L \wedge \sigma_k^L.\end{aligned} \tag{6.25}$$

and that the Lie brackets of the left and right vector fields are given by

$$\begin{aligned}[\xi_i^R, \xi_j^R] &= -\epsilon_{ij}{}^k\xi_k^R \\ [\xi_i^L, \xi_j^L] &= \epsilon_{ij}{}^k\xi_k^L \\ [\xi_i^R, \xi_j^L] &= 0.\end{aligned} \tag{6.26}$$

The last equation expresses the fact that the right (left) vector fields are left (right) invariant.

7 Conclusions and Outlook

In these lectures we have developed some of the basic tools needed to study duality in gauge theories with extended supersymmetry and have verified one non-trivial prediction of S duality. Of course this is a far cry from having a complete understanding of duality or even from testing it in a comprehensive way. At the time of writing this final section (March 1996) duality has turned into an enormous enterprise which is changing the way that we think about both field theory and string theory. I could not possibly summarize the current situation or the open problems and any attempt to do so would be obsolete within weeks. Instead let me end by mentioning some of the progress and open problems in the much more narrowly defined area of exact duality in $N = 4$ and finite $N = 2$ super Yang-Mills theory which has been the end of the logical development of these lectures.

1. There have been attempts to extend Sen's result to the full set of dyons states required by S duality [52, 53] but my impression is that no completely convincing construction yet exists.

2. All of the current tests of duality are really tests of the BPS spectrum of states or equivalently of states preserving half of the supersymmetry (this is true also of tests relying on topological field theory constructions). Yet if there is exact duality then it must relate all states and correlation functions including those at non-zero momentum. So far we do not have the tools to explore duality in a dynamical setting. For one partially successful attempt in this direction see [58].

3. As mentioned at the end of lecture 4, finite $N = 2$ theories are also conjectured to exhibit an exact duality symmetry. The simplest case involves gauge group $SU(2)$ with $N_f = 4$ hypermultiplets in the doublet representation of $SU(2)$. Some of the predictions of duality in this theory were explored in [6, 7].

4. In these lectures I have only considered duality with gauge group $SU(2)$. For larger gauge groups there are analogous predictions both for $N = 4$ Yang-Mills theory and for $N = 2$ theories with vanishing beta function. There has been recent progress in testing duality in these theories [59].

5. Finally, the central question remains of *why* these theories exhibit duality. It many cases it appears that duality in these theories is a low-energy manifestation of duality symmetries in string theory. The origin of duality in string theory is still an unsolved mystery.

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